

O-RAN Working Group 6 (Cloudification and Orchestration Workgroup) Acceleration Abstraction Layer High-PHY Profiles

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Foreword

This Technical Specification (TS) has been produced by WG6 of the O-RAN Alliance.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Executive summary

This document provides the description of the requirements and stage-2 information models to implement an Acceleration Abstraction Layer (AAL) for the High-PHY functionality hosted by an AAL-LPU that is associated with an O-RAN Distributed Unit (O-DU). Informative examples of stage-3 implementations are described in informational Annexes of this present document.

Introduction

This document specifies how to accelerate the High-PHY functionality of an O-DU where the High-PHY Acceleration Function (AF) is hosted on an AAL-LPU that is assigned to the AAL Application that is part of the O-DU. This document also specifies a set of AAL Profiles and procedures for utilizing the AALI Common Management (AALI-C-Mgmt) and Application (AALI-C-App) interfaces to convey the information elements specified by the AAL Profiles.

1 Scope

The present document specifies the requirements and stage-2 information models for the AAL_DOWNLINK_High-PHY and the AAL_UPLINK_High-PHY Profiles. Informative examples of stage-3 implementations for the AAL Profiles specified in this present document are described in informational Annexes of this present document.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] Void

[2] Void

[3] Voic

[4] 3GPP TS 38.211: "NR; Physical channels and modulation".

[5] 3GPP TS 38.212: "NR; Multiplexing and channel coding".

[6] 3GPP TR 38.300: "NR; NR and NG-RAN Overall Description".

[7] O-RAN WG6: Acceleration Abstraction Layer General Aspects and Principles, v06.00, November 2023

[8] O-RAN WG4 Control, User and Synchronization Plane Specification, v14.00, November 2023

[9] Void

[10] Void

[11] Acceleration Abstraction Layer AAL Profile - MUMIMO BeamForming/Precoder Calculation, v01.00, February 2020

[12] O-RAN WG6 Acceleration Abstraction Layer Common API v06.00, November 2023

2.2 Informative references

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

[i.1] SCF 222: "5G FAPI: PHY API Specification", v07.00, August 2023

[i.2] SCF 223: "5G FAPI: Front-End Unit API Specification", v05.00, August 2023

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3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in [7] apply.

3.2 Symbols

For the purposes of the present document, the symbols given in [7] apply.

3.3 Abbreviations

For the purposes of the present document, the terms and definitions given in the O-RAN Acceleration Abstraction Layer General Aspects and Principles [7] and the following abbreviations apply:

10	3GPP	3rd Generation Partnership Project
11	AALI	Acceleration Abstraction Layer interface
12	CB	Code Block
13	O-CU	O-RAN Centralized Unit
14	O-DU	O-RAN Distributed Unit
15	OFH	O-RAN Front Haul
16	O-RU	O-RAN Radio Unit
17	TB	Transport Block
18		

4 Overview

4.1 Purpose

The AAL General Aspects and Principles is described in [7] including a high level architecture of the AAL and definition of the AAL Profiles. This document details the AAL specification consisting of the requirements, information model and description of the interface to implement an AALI for the AAL_DOWNLINK_High-PHY and AAL_UPLINK_High-PHY profiles.

NOTE: The present document requires inline accelerators supporting High-PHY profiles in inline acceleration mode to implement both AAL_DOWNLINK_High-PHY and AAL_UPLINK_High-PHY profiles. Support for partial profiles is for FFS.

These profiles are illustrated below:

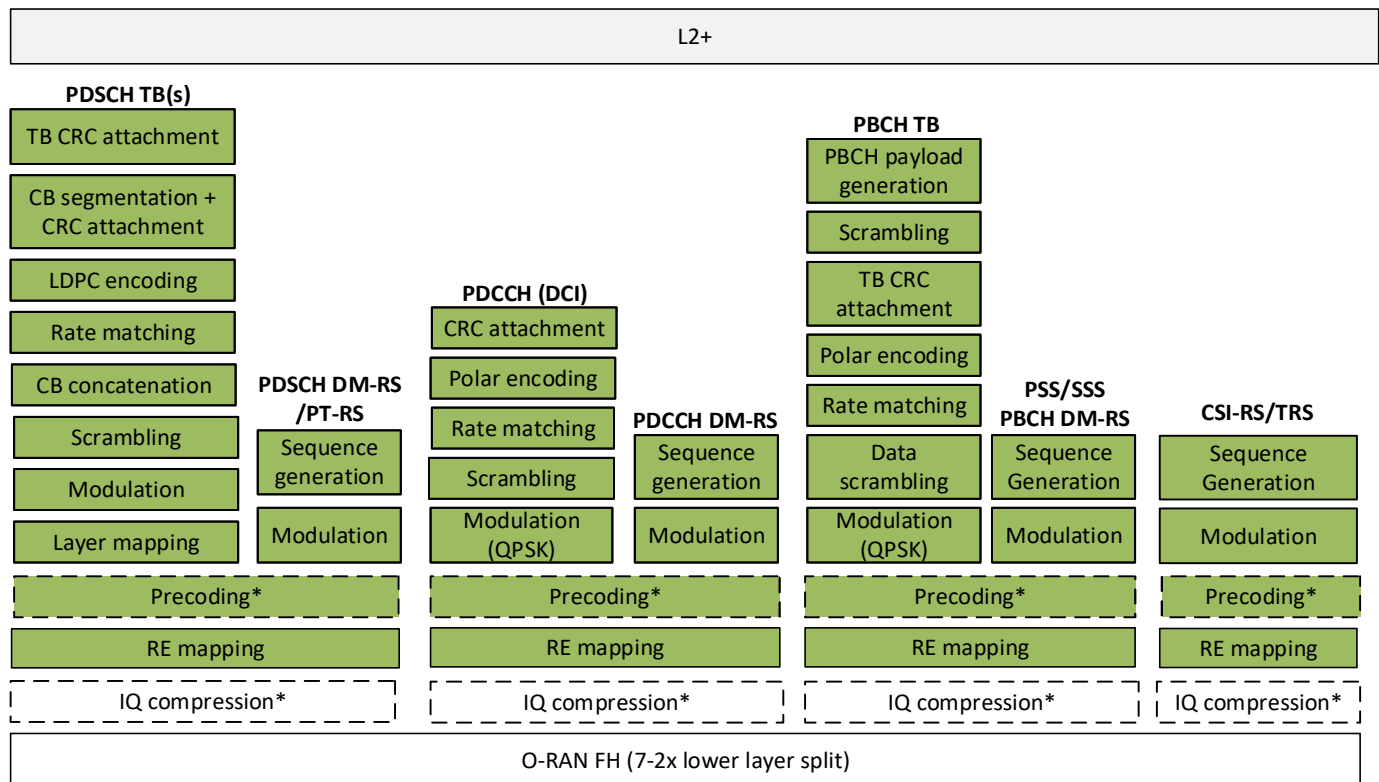


Figure 4.1-1: AAL_DOWNLINK_High-PHY Profile

5 AALI Configuration and Management

5.1 Introduction

The AALI-C-Mgmt APIs specified in [12] are used to configure the AAL Profiles specified in the present document. The AALI-C-App operations and events specified in [12] are used by the AAL Application to convey the AAL Profile's information elements specified in the present document.

5.2 Summary of Capabilities reported through the AALI-C-Mgmt Interface

Table 5.2-1 contains a list of capabilities for the AAL-Profiles specified in this present document. The reporting of these capabilities is optional and may be reported through the AALI-C-Mgmt interface as inputs to assist management systems in reconciling the configuration of the AAL-Profile with its capabilities along with any capability extensions. These capabilities are also used in the assignment of the AAL Application with an AAL-LPU supporting the AAL-Profiles specified in this present document.

Table 5.2-1: High-PHY Profile Capabilities Reported through the AALI-C-Mgmt Interface

Capability name	Data Type/Description	Properties	Applicable AAL-Profiles
carrierGroup	Data Type: array Description: The carrier groups supported by the AAL-Profile. A carrier is set of carriers that can pool capabilities (e.g., bandwidth layer).	x-support-qualifier: M x-isInvariant: True x-inventoryNotification True nullable: False x-isOrdered: False title: Numerologies minItems: 1 uniqueItems: True items: Numerology	AAL_DOWNLINK_High-PHY AAL_UPLINK_High-PHY

Table 5.2-2: Attribute Properties for Carrier Group

Attribute name	Data Type/Description	Properties
id	Data Type: string Description: The identifier of the carrier group.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification True nullable: False
supportedNumerologies	Data Type: array Description: The list of supported numerologies as described in table 4.2-1/3GPP TS 38.211 [4].	x-inventoryNotification True nullable: False format: integer8 minimum: 0 maximum: 4
numCarriers	Data Type: integer Description: Number of carriers for this carrier group.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification True nullable: False format: integer8 minimum: 1 maximum: 255
bandwidthLayers	Data Type: integer Description: The value is the aggregate bandwidth layers across all carriers in this carrier group.	x-support-qualifier: M readOnly: True x-isInvariant: True x-inventoryNotification True nullable: False

Attribute name	Data Type/Description	Properties
		format: integer64 x-units: HzLayer

6 AAL Profile Specifications

6.1 Profile Specifications Overview

This section contains information for each AAL Profile that is specified in this present document. An AAL Profile may provide APIs for a specific profile configuration that may be required in addition to the general AAL configuration APIs. AAL Profile specific configuration shall be done on a per AAL-LPU basis. Each AAL Profile shall define a list of capabilities that will be used to define the operations supported by the AAL-LPU.

A normative stage-3 definition for this High-PHY Profile specification is FFS and could leverage aspects of various options including but not limited to SCF-222.

An informative example of one possible mapping of the stage-2 HI-PHY Profile in this document is provided in Annex A.

6.1.1 Naming

Parameters and capabilities based on this stage-2 specification shall follow the following naming guidelines:

- Parameter and capability names shall start with a lowercase letter;
- No spaces may be used in the names;
- The names shall consist only of letters and digits and hyphens;
- The first letter of each non-initial word in a name shall be capitalized.
- Acronyms in the beginning of a parameter are lower case, uppercase otherwise
- Acronyms are hyphenated from preceding or succeeding upper case

6.2 O-DU AAL_DOWNLINK_High-PHY Profile Specification

This profile provides acceleration functionality for the following channels and signals:

- PDSCH (including Data, DM-RS and PT-RS)
- PDCCH (including Data and DM-RS)
- CSI-RS
- SSB (Including PSS, SSS and PBCH)

This clause presents parametrization for each of these channels and signals, and pertinent carrier configuration.

6.2.1 Profile Operation

The AAL_DOWNLINK_High-PHY Profile interface shall work on a slot basis, where the slot, numerology and SFN are signaled for the API.

The input data is specific to each downlink channel composing the profile. The output data is a set of C/U-Plane O-RAN Fronthaul (OFH) packets [8].

The AAL-LPU interfaces directly with the OFH and is responsible for handling its exchange of information with OFH.

As much as possible, parameter redundancy should be avoided. Where this is not possible, the AAL-LPU shall assume that redundant parameters are consistent.

6.2.2 Summary of Capabilities

The AAL_DOWNLINK_High-PHY Profile capabilities shall be reported to the application.

Table 6.2.2-1 lists a subset of the AAL_DOWNLINK_High-PHY profile capabilities that should be reported to the application with respect to the acceleration of the various channel functionality and interaction.

NOTE: Additional capabilities can also be reported by the AALI implementation.

AALI-P capapbi

Table 6.2.2-1: Sample AAL_DOWNLINK_High-PHY Profile Capabilities

Capability	Short summary	3GPP Spec Reference
3gpp-Release	Releases of 3GPP Specifications that are used. For example, Release 16, Release 17	https://www.3gpp.org/specifications/work-plan
maxPDSCH-TB-PerSlot	Number of PDSCH TB(s) per slot	Outside the scope of 3GPP
pdsch-CB-CRC	Supported with CB CRC caching in AAL-LPU, Supported with CB CRC reporting to L1, or Not supported	Outside the scope of 3GPP
cyclicPrefix	Normal, Extended or either	3GPP TS 38.211, sec 4.2
subcarrierSpacing-DL	15, 30, 60, 120, 240 kHz; can be channel-specific.	3GPP TS 38.211, sec 4.2
supportedBandwidth-DL	5, 10, 15, ... MHz	3GPP TS 38.104, sec 5.3
pdsch-MappingType	A, B or both	3GPP TS 38.211, sec 7.4.1.1.2
pdsch-AllocationType	0, 1 or both	3GPP TS 38.214, sec 5.1.2.2
pdsch-VRB-ToPRBMapping	interleaved, non-interleaved or both	3GPP TS 38.211, sec 7.3.1.6
pdsch-DMRS-ConfigTypes	1, 2 or both	3GPP TS 38.211, sec 7.4.1.1.2
pdsch-DMRS-AdditionalPos	pos0-pos3	3GPP TS 38.211, sec 6.4.1.1.3
pdsch-DMRS-MaxLength	1 or 2	3GPP TS 38.211, sec 7.4.1.1.2
pdsch-CBG	Supported with TB CRC caching in AAL-LPU, Supported with TB CRC reporting to the AAL Application, or Not supported	3GPP TS 38.212, sec 5.4.2.1, 7.3.1.2.2
maxModulationOrderDL	QPSK, 16-QAM, etc.	3GPP TS 38.214, sec 5.14.3
pdsch-PTRS	Supported or not	3GPP TS 38.211, sec 7.4.1.2
maxCORESETsPerSlot	Maximum number of CORESETs per slot	Outside the scope of 3GPP

cce-MappingType	Interleaved or not	3GPP TS 38.211, sec 7.3.2.2
precoderGranularityCORESET	Wideband (all contiguous RBs) or same as REG bundle	3GPP TS 38.211, sec 7.3.2.2
coreset-OutsideFirst3OfdmSymbols	Supported or not	3GPP TS 38.213, sec 13
pdccch-PrecoderSupport	Short Cyclic Delay Diversity, Precoder Cycling, None, etc.	Outside the scope of 3GPP
ssb-RateMatch	Rate Match around SSB: supported or not	
csi-RS-RM	Rate Match around CSI-RS: Supported or not	
rm-PatternTypeAndMethod	Rate Match around PrbSymbPattern: Bitmp, Coreset, Either or Neither	
pdccch-RateMatch	Rate Match around PDCCH: Supported or not	
lte-CRS-RM-Method	Rate Match around LTE CRS: Supported by static configuration, by explicit slot indication, not supported	
maxNumCarriersBW-LayersProduct DL	Maximum bandwidth*layers product support for forming joint precoders across bandwidth, and across baseband antenna ports. Includes the option of “no support”	Outside the scope of 3GPP
spatialMUX-PerDL-ChannelType-DL	Ability to spatially multiplex same or different channel types (PDSCH, PDCCH, etc.). Includes the option of “no support”	Outside the scope of 3GPP
maxNum-MU-MIMO-LayersPer-DL-ChannelType	The maximum number of layers that can be multiplexed (can be channel specific).	Outside the scope of 3GPP
maxNumber-MIMO-Layers-PDSCH	The maximum number of layers that can be allocated to a UE	3GPP TS 38.211, sec 7.3.1.3
precodingWeightsType	Support for using precoding weights supplied by the AAL Application, or computed in AAL-LPU [11], or semistatic look tables, or any combination.	Outside the scope of 3GPP
bch-PayloadFlag	Full payload (32 bits) or portion of payload originating from higher-layers (24 bits), or either	3GPP TS 38.212, sec 7.1.1
...	...	...

1

2 Aali-mgmtm

3 empty

6.2.2b Summary of Configuration

A carrier to which the AAL_DOWNLINK_High-PHY channel models apply shall be configurable by the application.

Table 6.2.2b-1 lists a subset of the carrier configurations pertinent to AAL_DOWNLINK_High-PHY.

NOTE: Additional configuration parameters are possible for the AALI implementation.

Table 6.2.2b-1: Sample Carrier Configurations pertinent to AAL_DOWNLINK_High-PHY

Configuration	Short summary	3GPP Spec Reference
dl-Bandwidth	Carrier bandwidth, for the DL transmission	3GPP TS 38.104, sec 5.2, 5.3 3GPP TS 38.211, sec 4.4.4.2
dl-Frequency	Carrier reference frequency (e.g. Point A, or RF reference frequency), for the DL transmission	3GPP TS 38.104, sec 5.2, 5.3 3GPP TS 38.211, sec 4.4.4.2
dl-GridSize	numerology μ , for DL transmission; can correspond to 15, 30, 60, 120, 240 kHz.	3GPP TS 38.211, sec 4.2
dl-CarrierOffset	Offset to carrier or k_0^μ , per numerology μ	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 5.3.1
num-TX-Ant	Number of baseband antenna ports for DL transmission	Outside the scope of 3GPP
prb-SymbRM-Patterns	Semistatic configuration of bitmap-based set of RBs and symbols, to reference for PDSCH rate match purposes	3GPP TS 38.214, sec 5.1.4.1 3GPP TS 38.331, sec 6.3.2
coreset-RM-Patterns	Semistatic CORESET-based set of RBs and symbols, to reference for PDSCH rate match purposes	3GPP TS 38.214, sec 5.1.4.1 3GPP TS 38.331, sec 6.3.2
lte-CRS-RM-Patterns	Semistatic LTE CRS patterns, to reference for PDSCH rate match purposes	3GPP TS 38.214, sec 5.1.4.1 3GPP TS 38.331, sec 6.3.2
beamformingTable	Semistatic configuration of beamforming vectors for DL transmission	Largely outside the scope of 3GPP. In some implementations, the beamforming vectors might be related to CSI-RS ports.
precodingTable	Semistatic configuration of precoding matrices for DL transmission	Largely outside the scope of 3GPP. In some implementations semistatic precoding matrices might be related to PMI reports in 3GPP TS 38.214, sec 5.2.2.2
[Semistatic configuration of DL Channel parameters.]	Alternative semistatic provisioning of some semistatic DL Channel parameters; See the respective Power Offset parameters in the tables in the quoted reference sections for e.g.:	c.f Table 6.2.6.2-3

	SSB: N_{ID}^{cell} , L_{max} , offsetToPointA, μ , μ_0 , k_{SSB} , Case, SSB candidates, SSB period, Power offset, beams – can be for multiple SSB configurations.	
[Power offset configurations]	Alternative semistatic configuration pertaining to semistatic power offset parameters for DL Channel models. See the respective Power offset parameters in the reference sections.	c.f Sections 6.2.3, 6.2.4, 6.2.5 and 6.2.6
[U-plane configuration]	Semistatic configuration for correct routing of OFH packets towards the O-RU. The parameters include, but not limited to: - Processing Elements - Configured low-level-tx-end points - Transmit carriers	Sec D.3.8 of ORAN-WG4-MP.0.v007.01

NOTE 1: Additional configuration parameters may be added in future releases to support additional features beyond basic 3GPP Rel-15.

NOTE 2: The summary description of configuration parameters outside the scope of 3GPP is not meant to restrict the mechanism or data type for such configuration.

6.2.3 PDSCH Channel Model

Per section 5.1.3.2.7 of the O-RAN AAL GAnP document [7] for the PDSCH High PHY Profile, the PDSCH Channel model of the AAL_DOWNLINK_High-PHY Profile supports acceleration of PDSCH Data, DM-RS and PT-RS functionalities.

The set of accelerated functions associated with the processing of PDSCH TB(s) is as follows:

- TB CRC attachment
- CB segmentation and CRC attachment
- LDPC encoding
- Rate matching
- CB concatenation
- Scrambling
- Modulation
- Layer mapping
- Precoding¹

¹ Configurable functional block, depends on implementation and/or system configuration

- RE mapping
- Power Offset
- IQ compression¹

The set of accelerated functions associated with the processing of PDSCH DM-RS is as follows:

- PDSCH DM-RS sequence generation
- Modulation
- Precoding¹
- RE mapping
- Power Offset
- IQ compression¹

The set of accelerated functions associated with the processing of PDSCH PT-RS is as follows:

- Sequence Generation
- Layer Mapping
- Modulation
- Precoding¹
- RE Mapping
- Power Offset
- IQ compression¹

6.2.3.1 PDSCH input and output for AAL_DOWNLINK_High-PHY Profile

The AAL_DOWNLINK_High-PHY profile shall signal PDSCH allocation(s) per slot. The input consists of the TB(s), and the associated parameters for the PDSCH allocation.

The output data shall consist of:

- in case of DL Transmission failure: asynchronous error indication with identifiers related to the corresponding PDSCH PDUs.

6.2.3.2 PDSCH Parameters

The following parameters are required to be supported by the AALI implementation when offloading operations. The AAL Application shall supply all relevant parameters; for ease of reading, the parameters are organized per signal type: Data, DM-RS, PT-RS.

PDSCH Data Parameters

Table 6.2.3.2-1: PDSCH Data Parameters

AAL-LPU parameters	Short summary	3GPP Spec Reference
--------------------	---------------	---------------------

MAC PDU for TB(s)		
nrOfCodewords	number of codewords in the PDSCH transmission (CW index $q = 0,1$) In case of single-codeword transmission, $q = 0$.	3GPP TS 38.214, sec 5.1.3.2 (max #CW) 3GPP TS 38.211, sec 7.3.1.1 (q index)
mac-PDU[q]	Initial transmission/non-CBG-based Re-tx: full TB to transmit CBG-based Re-tx: CBs to re-transmit	3GPP TS 38.214, sec 5.1.3.2 (non-CBG ReTx/initial Tx) 3GPP TS 38.214, sec 5.1.7 (CBG-based Tx)
tb-Size[q]	Transport block size. Computed at L2.	3GPP TS 38.214, sec 5.1.3.2
TB CRC		
dl-TB-CRC-CW[q]	Compute CRC once, reuse in case of re-transmission (CBG case: must be appended if last CB is retransmitted). NOTE: support for CRC caching is up to implementation and may be subject to capabilities.	3GPP TS 38.212, sec 7.2.1
CB Segmentation + CRC		
isLastCB-Present [q]	In case of CBG-based Re-tx, and last CBG is re-tx (CBGTI in 38.212), TB CRC is appended to the last CRC in the MAC Payload. Separate per CW CB CRC is computed in L1 accelerator from MAC PDU (no special controls needed for initial tx; see CB Concatenation section for handling options in case of CBG Re-tx) NOTE: Cached CRC provisioning is up to implementation and may be subject to capabilities.	3GPP TS 38.214, sec 5.1.3.2 3GPP TS 38.212, sec 7.3.1.2.2
LDPC Encoding		
rv-Index[q]	redundancy version, per CW. Determines starting position k_0 in the circular buffer. In DCI, or from aggregation, or from first TB.	3GPP TS 38.212, Table 5.4.2.1-2, 3GPP TS 38.214, Table 5.1.2.1-2
targetCodeRate[q] or (mcs-Table and mcs-Index[q]).	targetCodeRate[q]: target code rate (per CW) Explicit, for initial transmission. Implicit from tb-Size and allocation, for retransmissions. mcs-Table and mcs-Index[q]: target code rate can be extracted	3GPP TS 38.212, sec 5.4.2.1 3GPP TS 38.214, sec 5.1.3.1

	from the index (first transmission), or as above for retransmissions.	
newData[q]	Signals whether there is a new transmission, or a retransmission (per CW)	3GPP TS 38.212, sec 7.3.1.2.1/2
ldpc-Baseraph[q]	LDPC base graph to use (can be explicit, or derived from TB size and initial target rate) (per CW)	3GPP TS 38.212, sec 7.2.2
tb-SizeLBRM-Bytes or nCB	tbSizeLBRM-Bytes: Reference TBS for allocations subject to rateMatching = limitedBufferRM; impacts circular buffer length nCB = circular buffer length, after account for any FBRM/LBRM considerations.	3GPP TS 38.212, sec 5.4.2.1, 7.2.5
Rate Matching references (determining unavailable REs)		
ssb-PDUs-For RM	Set of SSBs, where PDSCH mapping is not possible.	3GPP TS 38.214, sec 5.1.4
prbSymRM-Patterns	Bitmap-based set of RBs and symbols not available for allocation	3GPP TS 38.214, sec 5.1.4.1 3GPP TS 38.331, sec 6.3.2
coreset-RM-Patterns	CORESET-based set of RBs and symbols not available for allocation	3GPP TS 38.214, sec 5.1.4.1 3GPP TS 38.331, sec 6.3.2
dci-Index	Index of a granting DCI; Rate Matching applies around the PDCCH grant index by this number, for the allocation (+ special consideration for AL16 candidates). See the 'same-index CCE Candidate' currently part of PDCCH Profile. Alternatively, that field can be signaled as a part of the PDSCH Profile, as it is applicable to PDSCH Rate Matching.	3GPP TS 38.214, sec 5.1.4.1
lte-CRS-RM-Patterns	LTE-CRS REs not available for allocation (+MBSFN awareness, for LTE-CRS RE mapping to symbols) NOTE: In 3GPP, this is an RRC parameter	3GPP TS 38.214, sec 5.1.4.2 3GPP TS 38.331, sec 6.3.2
csi-RS-For-RM	CSI-RS REs not available for allocation	3GPP TS 38.214, sec 5.1.4.2 3GPP TS 38.331, sec 6.3.2
CB Concatenation		

isLastCB-Present [q]		CBG-based Re-tx: presence of last CB requires TB CRC to be re-appended.	3GPP TS 38.214, sec 5.1.3.2 3GPP TS 38.212, sec 7.3.1.2.2
see section on MAC PDU for TB(s)		CB concatenation is handled in L2, in case of CBG-retx. CB CRC attachment in L1 Accelerator. Alternatively, CB concatenation can be handled exclusively in L1 accelerator, based on CBGTI signaling from L2	see section on MAC PDU for TB(s)
Scrambling			
nID-PDSCH or cInit [q]		nID-PDSCH: data scrambling identity (PCI by default) cInit[q]: scrambling initialization for codeword q.	3GPP TS 38.211, sec 7.3.1.1 3GPP TS 38.300, sec 3.1
rnti		RNTI associated with the PDSCH transmission	3GPP TS 38.211, sec 7.3.1.1
Modulation			
qam-ModOrder[q] or (mcs-Table and mcsIndex[q]).		qam-ModOrder[q]: Signaled via MCS in DCI; also impacts bit interleaving mcs-Table and mcs-Index[q]: modulation can be extracted from the MCS table and index.	3GPP TS 38.214, sec 5.1.3.1 3GPP TS 38.212, sec 5.4.2.2
Layer Mapping			
nrOfLayers		total number of layers (when > 4, assignment to codewords described in 38.211.	3GPP TS 38.211, sec 7.3.1.3 3GPP TS 38.214, sec 5.1
Precoding			
See section 6.2.7		Conceptually similar to FH signaling when precoding for Cat-B. 3GPP leaves DL precoding to implementation. Per layer (which also implies per CW)	
RE Mapping			
Frequency Domain	bwp-Start	Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PDSCH allocation	3GPP TS 38.211, sec 7.3.1.6
	bwp-Size	Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PDSCH allocation	3GPP TS 38.211, sec 7.3.1.6

	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	numDMRS-CDM-GrpsNoData	No PDSCH mapping on DMRS CDM groups marked as having no data.	3GPP TS 38.214, sec 5.1.3.2 3GPP TS 38.212, sec 7.3.1.2.2 3GPP TS 38.214, Table 4.1-1
	resourceAlloc	Mapping to VRBs: bitmap-based (type 0) or offset & length (type 1)	3GPP TS 38.214, sec 5.1.2.2
	rb-Bitmap	<u>type 0</u> : allocation is based on this bitmap. In 3GPP, it is signaled at RB group resolution	3GPP TS 38.214, sec 5.1.2.2.1 3GPP TS 38.212, sec 7.3.1.2.2
	rb-Start	<u>type 1</u> : start of allocation derived from DCI RIV	3GPP TS 38.214, sec 5.1.2.2.2 3GPP TS 38.212, sec 7.3.1.2.2
	rb-Size	<u>type 1</u> : number of contiguously allocated VRBs derived from DCI RIV	3GPP TS 38.214, sec 5.1.2.2.2 3GPP TS 38.212, sec 7.3.1.2.2
	vrB-ToPRB-Mapping	virtual resource blocks are mapped to physical resource blocks: interleaved or non-interleaved	3GPP TS 38.211, sec 7.3.1.6 3GPP TS 38.212, sec 7.3.1.2.1/2
	coreset-Type	RB bundling for VRB mapping depends on CORESET and DCI Type (BWP is assumed consistent with CORESET Type): “DCI 1_0 with SI-RNTI-based CRC in Type0-PDCCH CSS in CORESET0”, or otherwise.	3GPP TS 38.211, sec 7.3.1.6
Time Domain	startSymbolIndex	Start symbol index, for the allocation	3GPP TS 38.214, Table 5.1.2.1-1
	nrOfSymbols	Number of symbols, for the allocation	3GPP TS 38.214, Table 5.1.2.1-1
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation. NOTE: Applies only to $\mu=2$.	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
Rate Matching	<i>see the rate matching section</i>	Rate Matching determines which (RE x Symbol) resources are / are not available for RE mapping.	3GPP TS 38.214, sec 5.1.4
Power Offset			
	<acceleration based on hard-coded values>	Derived based on table 3GPP TS 38.214 table 4.1-1, DMRS Config Type and number of DMRS CDM groups without data	3GPP TS 38.214, sec 4.1, Table 4.1-1

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2 PDSCH DM-RS Parameters

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Table 6.2.3.2-2: PDSCH DM-RS Parameters

AAL-LPU parameters		Short summary	3GPP Spec Reference
Sequence Generation			
pdsch-DMRS-ScramblingId or cInitFirstSymbol		$N_{\{ID\}}$: scrambling identifier for $n_{\{SCID\}}$ $c_{\{init\}}[l]$: scrambling initialization for DMRS symbol.	3GPP TS 38.211, sec 7.4.1.1.1
n-SCID		sequence index (from DCI; defaults to 0)	3GPP TS 38.211, sec 7.4.1.1.1
Modulation			
<acceleration based on hard-coded values>		PDSCH DM-RS uses QPSK modulation	3GPP TS 38.211, sec 7.4.1.1.1
Precoding			
[see PDSCH data precoding]		DMRS follows PDSCH-data precoding	
RE Mapping			
Frequency Domain	bwp-Start	see PDSCH data table	3GPP TS 38.211, section 7.3.1.6
	bwp-Size	see PDSCH data table	3GPP TS 38.211, section 7.3.1.6
	subcarrierSpacing	see PDSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1
	resourceAlloc	see PDSCH data table	3GPP TS 38.214, sec 5.1.2.2
	rb-Bitmap	see PDSCH data table;e	3GPP TS 38.214, sec 5.1.2.2.1 3GPP TS 38.212, sec 7.3.1.2.2
	rb-Start	see PDSCH data table	3GPP TS 38.214, sec 5.1.2.2.2 3GPP TS 38.212, sec 7.3.1.2.2
	rb-Size	see PDSCH data table	3GPP TS 38.214, sec 5.1.2.2.2 3GPP TS 38.212, sec 7.3.1.2.2
	vr-b-to-PRB-Mapping	see PDSCH data table	3GPP TS 38.211, sec 7.3.1.6 3GPP TS 38.212, sec 7.3.1.2.1/2
	refPoint	NOTE: Reference point for RE mapping depends on CORESET Type and target for grant: - subcarrier 0 of lowest RB of CORESET0, or - subcarrier 0 of CRB0	3GPP TS 38.211, sec 7.4.1.1.2
	dmrs-Ports	Bitmap of DMRS ports, for each layer. OC weights applied to DM-RS REs are based on DM-RS CDM group	3GPP TS 38.211, sec 7.4.1.1.2

		and identifier for each DM-RS port.	
	dmrs-ConfigType	DMRS configuration type controls port to CDM group correspondence and frequency density (type 1: 8 ports, type 2:12 ports)	3GPP TS 38.211, sec 7.4.1.1.2
Time Domain	dl-DMRS-SymbPos	location of DMRS in slot sufficiently captures the impact of upper layer parameters, e.g., mapping type, additional pos	3GPP TS 38.211, sec 7.4.1.1.2 and Tables 7.4.1.1.2-3 and 7.4.1.1.2-4
	dmrs-Ports	Bitmap of DMRS ports, for each layer. OC weights applied to DM-RS REs are based on DM-RS CDM group and identifier for each DM-RS port.	3GPP TS 38.211, sec 7.4.1.1.2
	cyclicPrefix	see PDSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
Power Offset			
	powerControOffset-SS	power offset with respect to SSS	3GPP TS 38.214, sec 5.2.2.3.1 3GPP TS 38.214, sec 4.1 3GPP TS 38.213, sec 4.1

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2 PDSCH PT-RS Parameters

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Table 6.2.3.2-2: PDSCH PR-RS Parameters

AAL-LPU parameters	Short summary	3GPP Spec Reference
Sequence Generation		
ptrs-PortIndex	DMRS port associated with the PT-RS port according to clause 5.1.6.3 in [6, TS 38.214].	3GPP TS 38.211, sec 7.4.1.2.1
Layer Mapping		
ptrs-PortIndex	DMRS port associated with the PT-RS port according to clause 5.1.6.3 in [6, TS 38.214].	3GPP TS 38.211, sec 7.4.1.2.2 3GPP TS 38.214, sec 5.1.6.3
Modulation		
<acceleration based on hard-coded values>	PDSCH PTRS uses QPSK modulation, like PDSCH DMRS	3GPP TS 38.211, sec 7.4.1.1.1

Precoding			
<same as the associated dmrs Port(s)>			
RE Mapping			
Frequency Domain	bwp-Start	see PDSCH data table	3GPP TS 38.211, section 7.3.1.6
	bwp-Size	see PDSCH data table	3GPP TS 38.211, section 7.3.1.6
	subcarrierSpacing	see PDSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1
	ptrs-FreqDensity	frequency density (includes the case where there is no PT-RS)	3GPP TS 38.214, sec 5.1.6.3, Table 5.1.6.3-2
	ptrs-RE-Offset	re offset (can depend on DMRS association and RRC signaling of resourceElementOffset)	3GPP TS 38.211, sec 7.4.1.2.2 , table 7.4.1.2.2-1 3GPP TS 38.331, sec 6.3.1
	rnti	RNTI associated with the PDSCH transmission	3GPP TS 38.211, sec 7.4.1.2.2
Time Domain	ptrs-TimeDensity	time density (includes the case where there is no PT-RS)	3GPP TS 38.214, sec 5.1.6.3Table 5.1.6.3-1
	dl-DMRS-SymbPos	see PDSCH DMRS table symbols used for DMRS (PT-RS skips over these)	3GPP TS 38.211, sec 7.4.1.1.2 and Tables 7.4.1.1.2-3 and 7.4.1.1.2-4
	cyclicPrefix	see PDSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
Power Offset			
n-EPRE-RatioOf-PDSCH-To-PTRS		Derived based on table 3GPP TS 38.214 table 4.1-2, epre-Ratio and number of associated DMRS ports	3GPP TS 38.214, sec 4.1, Table 4.1-2

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6.2.4 PDCCH Channel Model

Per section 5.1.3.2.7 of the O-RAN AAL GAnP document [7] for the PDCCH High-PHY Profile, the PDCCH Channel model of the AAL_DOWNLINK_High-PHY Profile supports acceleration of PDCCH Data and DM-RS functionality.

The set of accelerated functions associated with the processing of PDCCH DCIs is as follows:

- CRC attachment
- Polar encoding
- Rate matching
- Scrambling
- Modulation (QPSK)
- Precoding¹

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- RE mapping
- IQ compression¹
- Power Offset

The set of accelerated functions associated with the processing of PDCCH DM-RS is as follows:

- Sequence generation
- Modulation
- Precoding¹
- RE mapping
- IQ compression¹

6.2.4.1 PDCCH input and output for AAL_DOWNLINK_High-PHY Profile

The AAL_DOWNLINK_High-PHY profile shall signal PDCCH CORESET allocation(s) per slot. The input data consists of the DCI payloads and the associated parameters for the PDCCH CORESET and DCI mapping.

The output data shall consist of:

- in case of DL Transmission failure: asynchronous error indication with identifiers related to the corresponding PDCCH PDUs.

6.2.4.2 PDCCH Parameters

The following parameters are required to be supported by the AALI implementation when offloading operations. The AAL Application shall supply all relevant parameters; for ease of reading, the parameters are organized per signal type: Data, DM-RS.

PDCCH Data Parameters

Table 6.2.4.2-1: PDCCH Data Parameters

AAL-LPU parameters		Summary	3GPP Spec Reference
MAC PDU(s) – DCI payloads			
num-DCI		number of DCIs in the CORESET	3GPP TS 38.213, section 10.1
per-DCI	payload	DCI payload generated by L2	3GPP TS 38.212, sec 7.3, 7.3.1
	payload-Size	Transport block size. Computed at L2.	3GPP TS 38.212, sec 7.3, 7.3.1
PDU CRC			

per-DCI	n-RNTI-PDCCH-Data	CRC computation detailed in 38.212 # payload size A from MAC # parity size L hardcoded to 24 bits # generator polynomial hard-coded to $g_{\{CRC24C\}}(D)$ # CRC scrambled with RNTI	3GPP TS 38.212, sec 7.3.2
Polar Encoding			
aggregationLevel		Encoding detailed in 38.212: # input sequence (payload with CRC) has $K=A + 24$ bits # $n_{\{max\}}$ is hardcoded to 9 # $I_{\{IL\}}$ is hardcoded to 1 # $n_{\{PC\}}$ is hardcoded to 0 # $n_{\{PC\}}^{\{wm\}}$ is hardcoded to 0 =====	3GPP TS 38.212, sec 7.3.3
		N: encoded bit length is a direct function of K (A+24) and E (#non-DMRS PDCCH symbols based on $AL * 2$) and a set of hardcoded parameters. Interleaving: function of I_{IL} (hardcoded for PDCCH) and a set of additional hardcoded table and parameters, operates on the payload Encoding: function of $n_{\{PC\}}$ (hardcoded for PDCCH), $n_{\{PC\}}^{\{wm\}}$ (hardcoded for PDCCH), + additional spec hard-coded tables and parameters, operates on the interleaved payload	
Rate Matching			
aggregationLevel		Rate matching detailed in 38.212 # out size $E = 2 * \#non-DMRS \text{ REs}$ (function of Aggregation Level) # rate match algorithm uses this E, the size of the original sequence + CRC (K), $n_{\{PC\}}$ (hardcoded to 0) and $n_{\{BIL\}}$ (hardcoded to 0) to operate on the polar-coded payload	3GPP TS 38.214, sec 7.3.4
Scrambling			
per-DCI	n-ID-PDCCH-Data or cInit	$n_{\{ID\}}$: data scrambling identity (PCI by default, can be UE-specific) $c_{\{init\}}$: scrambling initialization	3GPP TS 38.211, sec 7.3.2.3

	n-RNTI-PDCCH-Data	RNTI-based scrambling associated with the DCI transmission (0 for CSS)	3GPP TS 38.211, sec 7.3.2.3
Modulation			
	<acceleration based on hard-coded values>	PDCCH uses QPSK modulation	3GPP TS 38.211, sec 7.3.2.4 3GPP TS 38.211, sec 5.1.3
Precoding			
per-DCI	See section 6.2.7	Conceptually similar to FH signaling when precoding for Cat-B. 3GPP leaves DL precoding to implementation.	
RE Mapping			
Frequency Domain	coreset-BWP-Start	Start, w.r.t. CRB, of bandwidth part: RBs indexing for the frequency domain allocation is relative to the Bandwidth part for the PDCCH allocation	3GPP TS 38.211, sec 7.3.2.2 3GPP TS 38.331, sec 6.3.2
	coreset-BWP-Size	Size of bandwidth part: RBs indexing for the frequency domain allocation is relative to the Bandwidth part for the PDCCH allocation	3GPP TS 38.211, section 7.3.1.6
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	regBundleSize	Number of REGs in an REG Bundle	3GPP TS 38.211, sec 7.3.2.2 3GPP TS 38.331, sec 6.3.2
	interleaverSize	Interleaver size	3GPP TS 38.214, sec 5.1.2.2.2 3GPP TS 38.212, sec 7.3.1.2.2
	cce-REG-MappingType	Mapping of Control Channel Elements (CCE) to Resource Element Groups (REG)	3GPP TS 38.211, sec 7.3.2.2 3GPP TS 38.211, sec 7.4.1.3.2 3GPP TS 38.331, sec 6.3.2
	freqDomainResource	Bitmap signaling the frequency domain resources for the CORESET. Each bit corresponds a group of 6 RBs, with grouping starting from the first RB group in the BWP. Validity and bit mapping per the RRC parameter frequencyDomainResources in 38.331 This parameter applies to both regular CORESETs, as well as CORESET0. For the interpretation regarding CORESET0, see section 13 of 38.213.	3GPP TS 38.211, sec. 7.3.2.2 3GPP TS 38.331, sec 6.3.2 3GPP TS 38.213, sec 13

	shiftIndex	offset used in the computation of the interleaver function for CCE-to-REG mapping	3GPP TS 38.211, sec 7.3.2.2
	precoderGranularity	Precoder granularity in frequency domain (for data, this is strictly not needed, as it will be accounted for in the precoding)	3GPP TS 38.211, sec 7.3.2.2 3GPP TS 38.211, sec 7.4.1.3.2
FD per DCI	cce-Index	cce Index for sending the DCI	3GPP TS 38.213, sec 10.1
	collocated-AL16-Candidate	indicates presence of an AL-16 candidate existing at the same CCE Index, for PDSCH rate matching purposes NOTE: This information is relevant to PDSCH rate matching, and could be signaled in PDSCH for rate matching purposes, instead. See the Rate Matching section of the PDSCH Profile	3GPP TS 38.214, sec 5.1.4.1
	aggregationLevel	aggregation level for the DCI	3GPP TS 38.211, sec 7.3.2.1 3GPP TS 38.213, sec 10.1
Time Domain	startSymboIndex	Start symbol index, for the allocation	3GPP TS 38.211, sec. 7.3.2.2
	durationSymbols	Number of symbols, for the allocation	3GPP TS 38.211, sec. 7.3.2.2
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
Power Offset			
[Power offset w.r.t. SSS] can be hardcoded, for MVP.		see PDCCH DMRS table. Can be set to the same value.	3GPP TS 38.214, sec 5.2.2.3.1 3GPP TS 38.214, sec 4.1 3GPP TS 38.213, sec 4.1

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2 PDCCH DM-RS Parameters

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Table 6.2.4.2-2: PDCCH DM-RS Parameters

AAL-LPU parameters		Summary	3GPP Spec Reference
per DCI	n-ID-PDCCH-Data or cInit	N_{ID}: scrambling identifier c_{init}: scrambling initialization for DMRS.	3GPP TS 38.211, sec 7.4.1.3.1
Modulation			
<acceleration based on hard-coded values>		PDCCH DM-RS uses QPSK modulation	3GPP TS 38.211, sec 7.4.1.3.1
Precoding			

per DCI	[see PDCCH data precoding]	DMRS follows PDCCH-data precoding	
RE Mapping			
Frequency Domain	coreset-BWP-Start	see PDCCH data table	3GPP TS 38.211, section 7.3.2.2
	coreset-BWP-Size	see PDCCH data table	3GPP TS 38.211, section 7.3.2.2
	subcarrierSpacing	see PDCCH data table	3GPP TS 38.211, sec 4.2, 5.3.1
	coreset-Type	reference point for RE mapping physical resources of DMRS depends on CORESET type (#0 or not)	3GPP TS 38.211, sec 7.4.1.3.2
	precoderGranularity	Precoder granularity in frequency domain (impacts whether DMRS should be generated for all RBs or only allocated RBs)	3GPP TS 38.211, sec 7.3.2.2 3GPP TS 38.211, sec 7.4.1.3.2
Freq Domain per DCI	cce-Index	see PDCCH data table	3GPP TS 38.213, sec 10.1
	aggregationLevel	see PDCCH data table	3GPP TS 38.211, sec 7.3.2.1 3GPP TS 38.213, sec 10.1
Time Domain	startSymbolIndex	see PDCCH data table	3GPP TS 38.211, sec. 7.3.2.2
	durationSymbols	see PDCCH data table	3GPP TS 38.211, sec. 7.3.2.2
	cyclicPrefix	see PDCCH data table	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
Power Offset			
powerControlOffsetSS		power offset with respect to SSS	3GPP TS 38.214, sec 5.2.2.3.1 3GPP TS 38.214, sec 4.1 3GPP TS 38.213, sec 4.1

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6.2.5 CSI-RS Channel Model

Per section 5.1.3.2.7 of the O-RAN AAL GAnP document [7] for the CSI-RS High PHY Profile, the CSI-RS Channel model of the AAL_DOWNLINK_High-PHY Profile supports acceleration of CSI-RS functionality.

The set of accelerated functions associated with the processing of CSI-RS is as follows:

- CSI-RS sequence generation
- Modulation
- Precoding¹
- RE mapping
- IQ compression¹
- Power Offset

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6.2.5.1 CSI-RS input for AAL_DOWNLINK_High-PHY Profile

The AAL_DOWNLINK_High-PHY profile shall signal CSI-RS Resource(s) per slot. The input consists of the CSI-RS resource parameters.

CSI-RS Parameters

The following parameters are required to be supported by the AALI implementation when offloading operations. The AAL Application shall supply all relevant parameters.

6.2.5.2 CSI-RS Parameters

Table 6.2.5.2: CSI-RS Parameters

AAL-LPU Parameters		Summary	3GPP Spec Reference
Sequence Generation			
scrambId or cInitFirstSymbol		n_{ID}: scrambling id c_{init}[l]: scrambling initialization for symb [l]	3GPP TS 38.211, sec 7.4.1.5.2
Modulation			
<acceleration based on hard-coded values>		CSI-RS uses QPSK modulation	3GPP TS 38.211, sec 7.4.1.5.2
Precoding			
See section 6.2.7		Conceptually similar to FH signaling when precoding for Cat-B. 3GPP leaves DL precoding to implementation.	
RE Mapping			
Frequency Domain	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	startingRB	PRB where this CSI resource starts in relation to CRB#0	3GPP TS 38.331, sec 6.3.2
	nrOfRBs	Number of PRBs across which this CSI resource spans.	3GPP TS 38.331, sec 6.3.2
	freqDomain	Bitmap defining the frequencyDomainAllocation, with interpretation subject to the Row selection for table 7.4.1.5.3-1	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 7.4.1.5.3
	row	row indicating the CSI-RS location in table 7.4.1.5.3-1. Can be used to derive; density ρ (see also Frequency Density)	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 7.4.1.5.3

		• cdmType • ports • $\bar{k}$ in the $(\bar{k}, \bar{l})$-tuple	
	freqDensity	Indicates whether even or odd PRBs are occupied by CSI-RS. Applicable when density = dot5 (0.5)	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 7.4.1.5.3
Time Domain	row	row indicating the CSI-RS location in table 7.4.1.5.3-1. Can be used to derive; • $\bar{l}$ in the $(\bar{k}, \bar{l})$-tuple	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 7.4.1.5.3
	symbL0	value of l_0 for the interpretation of table 7.4.1.5.3-1. Signaled by RRC parameter firstOFDMSymbolInTimeDomain	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 7.4.1.5.3
	symbL1	value of l_1 for the interpretation of table 7.4.1.5.3-1. Signaled by RRC parameter firstOFDMSymbolInTimeDomain2	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 7.4.1.5.3
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
Power Offset			
	powerControlOffsetSS	power offset with respect to SSS	3GPP TS 38.214, sec 5.2.2.3.1

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6.2.6 SSB Model

Per section 5.1.3.2.7 of the O-RAN AAL GAnP document [7] for the PBCH High-PHY Profile, the SSB Channel model of the AAL_DOWNLINK_High-PHY Profile supports acceleration of PSS+SSS and PBCH Data and PBCH DM-RS functionality.

The set of accelerated functions associated with the processing of PSS+SSS is as follows:

- Sequence generation
- Modulation
- Precoding¹
- RE mapping
- IQ compression¹
- Power Offset

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The set of accelerated functions associated with the processing of PBCH Data is as follows:

- PBCH payload generation
- Scrambling
- TB CRC attachment
- Polar encoding
- Rate matching
- Data scrambling
- Modulation (QPSK)
- Precoding¹
- RE mapping
- IQ compression¹

The set of accelerated functions associated with the processing of PBCH DM-RS is as follows:

- Sequence generation
- Modulation
- Precoding¹
- RE mapping
- IQ compression¹
- Power Offset

6.2.6.1 SSB input for AAL_DOWNLINK_High-PHY Profile

The AAL_DOWNLINK_High-PHY profile shall signal SSB allocation(s), per slot (up to two per slot and SSB configuration, at the SSB slot numerology). The input consists of the PBCH payload and the associated SSB resource allocation parameters.

6.2.6.2 SSB Parameters

The following parameters are required to be supported by the AAL implementation when offloading operations. The AAL Application shall supply all relevant parameters; for ease of reading, the parameters are organized per signal type: PBCH Data, PBCH DM-RS, PSS&SSS.

PBCH Data Parameters

Table 6.2.6.2-1: PBCH Data Parameters

AAL-LPU Parameters	Summary	3GPP Spec Reference
MAC PDU(s)		
bch-Payload	PBCH payload (generated by L2, or in AAL-LPU) - may be the full payload (32 bits) or only the portion supplied from higher layers (24 bits).	3GPP TS 38.212, sec 7.1, 7.1.1

Payload Scrambling		
physCellId or cInit	$N_{\{ID\}^{\{cell\}}}$: Physical layer cell ID, as defined in as defined in 3GPP TS 38.211, section 7.4.2.1 $c_{\{init\}}$: scrambling initialization	3GPP TS 38.211, sec 7.4.1.4.1, 7.1.2
l-Max	Maximum number of candidate SS/PBCH blocks in a half frame	3GPP TS 38.211, sec 7.3.3.1, 7.1.2
CRC		
<acceleration based on hard-coded values>	CRC computation detailed in 38.212: # (scrambled) payload size A from MAC: $24+8=32$ bits (Rel-15,16) # parity size L hardcoded to 24 bits # generator polynomial hard-coded to $g_{\{CRC24C\}}(D)$	3GPP TS 38.212, sec 7.1.3
Polar Coding		
<acceleration based on hard-coded values>	Encoding detailed in 38.212: # input sequence (scrambled payload with CRC) has $K = B = 32 + 24 = 56$ bits (result of adding up two hardcoded numbers) # $n_{\{max\}}$ is hardcoded to 9 # $I_{\{IL\}}$ is hardcoded to 1 # $n_{\{PC\}}$ is hardcoded to 0 # $n_{\{PC\}}^{\{wm\}}$ is hardcoded to 0 ===== N: encoded bit length is a direct function of K (56) and E (rate match output 864, spelled out in spec) and a set of hard-coded parameters = 512 Interleaving: function of $I_{\{IL\}}$ (hardcoded for PBCH) and a set of additional hardcoded table and parameters, operates on the payload Encoding: function of $n_{\{PC\}}$ (hardcoded for PBCH), $n_{\{PC\}}^{\{wm\}}$ (hardcoded for PBCH), + additional spec hard-coded tables and parameters, operates on the interleaved payload	3GPP TS 38.212, sec 7.1.4

Rate Matching			
<acceleration based on hard-coded values>		Rate matching detailed in 38.212 # out size $E = 864$, explicit in the spec, but can be derived as (#non-DMRS REs * 2) # rate match algorithm uses this E , the size of the original sequence + CRC ($K = 56$), $n_{\{PC\}}$ (hardcoded to 0) and $n_{\{BIL\}}$ (hardcoded to 0) to operate on the polar-coded payload	3GPP TS 38.212, sec 7.1.5
Scrambling			
l-Max		Maximum number of candidate SS/PBCH blocks in a half frame	3GPP TS 38.211, sec 7.3.3.1
ssb-BlockIndex		candidate SS/PBCH block index	3GPP TS 38.211, sec 7.3.3.1
Modulation			
<acceleration based on hard-coded values>		PBCH data uses QPSK modulation	3GPP TS 38.211, sec 7.3.3.2 3GPP TS 38.211, sec 5.1.3
Precoding			
See section 6.2.7		Conceptually similar to FH signaling when precoding for Cat-B. 3GPP leaves DL precoding to implementation.	
RE Mapping			
Frequency Domain	ssb-OffsetToPointA	Frequency offset between point A and the lowest subcarrier of the lowest resource block, which has the subcarrier spacing provided by the higher-layer parameter subCarrierSpacingCommon and overlaps with the SS/PBCH block"	3GPP TS 38.211, sec 4.4.4.2
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	ssb-SubcarrierOffset	subcarrier offset	3GPP TS 38.211, sec 7.4.3.1
Time Domain	case	Case determines first symbol mapping	3GPP TS 38.213, sec 4.1
	ssb-BlockIndex	Used to derive the SSB index in slot	3GPP TS 38.213, sec 4.1
Power Offset			
<acceleration based on hard-coded values>		The UE assumes that SSS, PBCH DM-RS, and PBCH data have same EPRE	3GPP TS 38.213, sec 4.1

PBCH DM-RS Parameters

Table 6.2.6.2-2: PBCH DM-RS Parameters

AAL-LPU Parameters		Summary	3GPP Spec Reference
Sequence Generation			
physCellId		Physical cell ID	3GPP TS 38.211, sec 7.4.1.4.1, 7.4.2.1
l-Max		Maximum number of candidate SS/PBCH blocks in a half frame	3GPP TS 38.211, sec 7.4.1.4.1
i-SSB		candidate SSB SS/PBCH block index; can also be derived from l-Max and ssb-BlockIndex	3GPP TS 38.211, sec 7.4.1.4.1
Modulation			
<acceleration based on hard-coded values>		PBCH DM-RS uses QPSK modulation	3GPP TS 38.211, sec 7.4.1.4.1
Precoding			
[see PBCH data precoding]		DMRS follows PDCCH-data precoding	
RE Mapping			
Frequency Domain	offsetToPointA	see PBCH data	3GPP TS 38.211, sec 4.4.4.2
	subcarrierSpacing	see PBCH data	3GPP TS 38.211, sec 4.2, 5.3.1
	ssb-SubcarrierOffset	see PBCH data	3GPP TS 38.211, sec 7.4.3.1
Time Domain	case	see PBCH data	3GPP TS 38.213, sec 4.1
	ssb-BlockIndex	see PBCH data	3GPP TS 38.213, sec 4.1
Power Offset			
<acceleration based on hard-coded values>		The UE assumes that SSS, PBCH DM-RS, and PBCH data have same EPRE	3GPP TS 38.213, sec 4.1

PSS & SSS Parameters

Table 6.2.6.2-3: PSS & SSS Parameters

AAL-LPU Parameters		Summary	3GPP Spec Reference
Sequence Generation			

physCellId		Physical cell ID - used to derive N_ID(1) for SSS and N_ID(2) for PSS	3GPP TS 38.211, sec 7.4.2.1, 7.4.2.2.1, 7.4.2.3.1
Modulation			
<acceleration based on hard-coded values>		PSS and SSS use BPSK modulation	3GPP TS 38.211, sec 7.4.2.2.1, 7.4.2.3.1
Precoding			
See section 6.2.7		Conceptually similar to FH signaling when precoding for Cat-B. 3GPP leaves DL precoding to implementation.	
RE Mapping			
Freq Domain	offsetToPointA	see PBCH data tab	3GPP TS 38.211, sec 4.4.4.2
	subcarrierSpacing	see PBCH data tab	3GPP TS 38.211, sec 4.2, 5.3.1
	ssb-SubcarrierOffset	see PBCH data tab	3GPP TS 38.211, sec 7.4.3.1
Time Domain	case	see PBCH data tab	3GPP TS 38.213, sec 4.1
	ssb-BlockIndex	see PBCH data tab	3GPP TS 38.213, sec 4.1
Power Offset			
beta-PSS		PSS EPRE to SSS EPRE in an SSB	3GPP TS 38.213, sec 4.1

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6.2.7 Beamforming

In this release, the AAL_DOWNLINK_High-PHY profile supports the following beamforming methods available for OFH signaling:

- Predefined-Beam Beamforming, as defined in section 10.4.2.1 of [8];
- Weight-based Dynamic Beamforming, as defined in section 10.4.2.2 of [8];
- Attribute-Based Dynamic Beamforming, as defined in section 10.4.3 of [8];
- Channel-Information-Based Beamforming, as defined in section 10.4.4 of [8].

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6.2.7.1 Predefined-Beam Beamforming

The AALI shall support the AAL Application signaling of the following parameters, as needed for AAL-LPU-signaling of beamId in C-Plane Section Types 1, 3 [8]:

- frequency-domain beam indices
- time-domain-domain beam indices
- a mixture of the two (“hybrid beamforming”).

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6.2.7.2 Weight-based Dynamic Beamforming

The AALI shall support the AAL Application signaling of the following parameters, as needed for AAL-LPU-signaling of beamforming weight vectors of (bfwI, bfwQ) in C-Plane Extensions 1, 11, 19 [8]:

Table 6.2.7.2-1: Weight-based Dynamic Beamforming Parameters for Downlink

AAL-LPU Parameters	Summary	AAL-LPU Role
AAL-LPU → AAL Application		
Channel Estimation Abstraction	A representation of SRS-based channel observations, as documented in the SRS channel model in section 6.3.5.2, e.g. for usage antennaSwitch	Signals an abstraction of the Channel Estimation to the AAL Application
AAL Application → AAL-LPU		
rnti[*]	Selected UEs for scheduling (for an illustration refer to the L users referenced in Appendix J.4 of [8])	Compute precoding weights for the selected UEs and layers, e.g. based on the reported Channel Estimation and its Abstraction (if beamforming weights are not provided by the AAL Application).
chosenLayers[*]	Per UE: selected Layers for scheduling (for an illustration refer to the K layers referenced in Appendix J.4 of [8])	
Beamforming weights, per UE [optional]	Beamforming weights for the layers of the selected UEs, where beamforming weights may be specified explicitly or may be sent to AAL-LPU as handle to AAL-LPU [11] contexts that uniquely specify the beamforming weights to apply.	Signal to O-RU beamforming vectors of weights (bfwI, bfwQ) over C-Plane for the AAL Application-selected UEs and layers. Alternatively, the beamforming weights may be consumed in AAL-LPU itself, e.g. Cat-A, when precoding is applied in the O-DU.

6.2.7.3 Attribute-Based Dynamic Beamforming

The AALI shall support the AAL Application signaling of the following parameters, as needed for AAL-LPU-signaling of beamforming attributes (bfAzPt, bfZePt, bfAz3dd, bfZe3dd, bfAzSl, bfZeSl) in C-Plane Extensions 2 [8]:

- Zenith main and 3dB angles
- Azimuth main and 3dB angles
- Sidelobe Angles

6.2.7.4 Channel-Information-Based Beamforming

The AALI shall support the AAL Application signaling of the following parameters, as needed for AAL_DOWNLINK_High-PHYprofile signaling of channel estimate vectors of (ciIsample, ciQsample) in C-Plane Section 6 [8]:

Table 6.2.7.4-1: Channel-Information-Based Beamforming Parameters for Downlink

AAL-LPU Parameters	Summary	AAL-LPU Role
AAL-LPU → AAL Application		
[Channel Estimation Abstraction]	A representation of SRS-based channel observations, as documented in the SRS channel model in section 6.3.5.2, e.g. for usage antennaSwitch	Signals an abstraction of the Channel Estimation to the AAL Application
AAL Application → AAL-LPU		
rnti[*]	Selected UEs for scheduling (for an illustration refer to the L users referenced in Appendix J.4 of [8])	Signal to O-RU the Channel Estimates for the AAL Application-selected UEs and layers
chosenLayers[*]	Per UE: selected Layers for scheduling (for an illustration refer to the K layers referenced in Appendix J.4 of [8])	

6.3 O-DU AAL_UPLINK_High-PHY Profile Specification

This profile provides acceleration functionality for the following channels and signals:

- PUSCH (including Data, DM-RS and PT-RS)
- PUCCH (Data) – Format 0
- PUCCH (including Data and DM-RS) – Format 1
- PUCCH (including Data and DM-RS) – Formats 2, 3, 4
- SRS
- PRACH

This section presents parametrization for each of these channels and signals, and pertinent carrier configuration.

6.3.1 Profile Operation

The AAL_UPLINK_High-PHY Profile interface shall work on a slot basis. An operation can be performed one slot at a time, where the slot, numerology and SFN are signaled for the API.

The output data is specific to each uplink channel composing the profile.

The input data consists of:

- a set of AAL Application-supplied parameters for each of the modelled uplink channels
- a set of U-Plane OFH packets.

For the OFH packets, the AAL-LPU interfaces directly with the OFH and is responsible for handling its exchange of information with OFH.

As much as possible, parameter redundancy should be avoided. Where this is not possible, the AAL-LPU can assume that redundant parameters are consistent.

6.3.2 Summary of Capabilities

The AAL_UPLINK_High-PHY Profile capabilities shall be reported to the application.

Table 6.3.2-1 lists a subset of the AAL_UPLINK_High-PHY profile capabilities that should be reported to the application with respect the acceleration of the various channel functionalities and interaction.

NOTE: Additional capabilities can also be reported by the AALI implementation.

Table 6.3.2-1: Sample AAL_UPLINK_High-PHY Profile Capabilities

Capability	Short summary	3GPP Spec Reference
3gpp-Release	Rel-15, Rel-16, ...	https://www.3gpp.org/specifications/work-plan
max-PUSCH-TB-PerSlot	Number of PUSCH TBs per slot	Outside the scope of 3GPP
cyclicPrefix	Normal or Extended	3GPP TS 38.211, sec 4.2
subcarrierSpacing-UL	15, 30, 60, 120, 240 kHz; can be channel-specific.	3GPP TS 38.211, sec 4.2
supportedBandwidth-UL	5, 10, 15, ... MHz	3GPP TS 38.104, sec 5.3
uci-MUX-UL-SCH-in-PUSCH	Supported or not	3GPP TS 38.212, sec 6.3.2
pusch-FrequencyHopping	Supported or not	3GPP TS 38.214, sec 6.3
pusch-DMRS-ConfigTypes	1 or 2 or both	3GPP TS 38.211, sec 6.4.1.1.3
pusch-DMRS-MaxLen	1 or 2	3GPP TS 38.211, sec 6.4.1.1.3
pusch-MappingType	A or B or both	3GPP TS 38.211, sec 6.4.1.1.3
dfts-OFDM	Supported or not	3GPP TS 38.211, sec 6.3.1.4
pusch-CBG	Supported or not	3GPP TS 38.212, sec 5.4.2.1, 7.3.1.1.2
pusch-Max-PTRS	Maximum number of UL PTRS	3GPP TS 38.211, sec 6.4.1.2
pusch-AggregationFactor	1-8 or not supported	3GPP TS 38.214, sec 6.1.2.1
pusch-LBRM-Support	Supported or not	3GPP TS 38.212, sec 5.4.2.1
pucch-Formats	Bitmap indicating the support for Formats 0, 1, 2, 3 or 4	3GPP TS 38.211, sec 6.3.2
pucch-GroupAndSequenceHopping	Group, sequence or neither	3GPP TS 38.211, sec 6.3.2.2
max-PUCCH-PerSlot	Max number of PUCCH Resources per slot	Outside the scope of 3GPP
pucch-Aggregation	Supported or not (formats 1, 3, 4)	
supported-SRS-Usage	beamManagement, codebook, nonCodebook, antennaSwitching, including any combination. If none of the supported reports (see “SRS report type support”) requires usage	3GPP TS 38.214, sec 6.2.1

	awareness, this usage support may be omitted or set to all usage types.	
allowed-SRS-reportsPerUsage	e.g. SVD (singular value decomposition), or H (raw channel) report or SINR, or 2D-DFT basis, or other reports, subsampled in frequency domain.	Sec 0
dft-Size-Azimuth and dft-Size-Elevation	DFT size and oversampling factor in azimuth and in elevation (only applicable when SRS report type 2D-DFT basis is supported). The DFT size in azimuth should equal the product of the number of antenna elements in azimuth and the azimuth oversampling factor. The DFT size in elevation should equal the product of the number of antenna elements in elevation and the elevation oversampling factor.	Outside the scope of 3GPP
maxNumStrongestBeamsToReport	Parameter K_DFT_beams defined as the number of strongest beams per SRS basis vector for SRS report type 2D-DFT basis (only applicable when SRS report type 2D-DFT basis is supported).	Outside the scope of 3GPP
min-RB-resolutionFor-SRS	RB resolution at which the SRS reports may be subsampled	Outside the scope of 3GPP
max-SRS-ChannelReportsPerSlot	Maximum number of SRS reports per slot	Outside the scope of 3GPP
max-SRS-TX-AntennaPortsPerUE	Maximum number of ports to sample per UE	Partly outside the scope of 3GPP 3GPP TS 38.211, sec 6.4.1.4.1
SRS Configurations: - maxNumConsecutive-SRS-Slots - srs-FrequencyHopping - max--SRS-CombSize - maxNumSRS-CyclicShifts	Capabilities regarding SRS configurations: • Support for consecutive SRS symbols • SRS frequency hopping • Comb Size • Cyclic shifts	3GPP TS 38.211, sec 6.4.1.4
SRS distribution: - maxDutyCycleInPercentage - srs-SymbolsBitmap - maxNumSRS-SymbPerSlot	Capabilities regarding SRS occurrence in time: • Duty Cycle: Maximum proportion of SRS-bearing slots in an SRS period. • Bitmap of symbol locations per slot	3GPP TS 38.211, sec 6.4.1.4 Partly outside 3GPP Scope

	Symbols per slot This capability is agnostic to SRS trigger type (e.g. periodic / semi-persistent / aperiodic).	
prach-LongFormats prach-ShortFormats	0-3, A1-3, B1-4, C0-1	3GPP TS 38.211, sec 6.3.3
prach-RestrictedSets	Type A, B, none	3GPP TS 38.211, sec 6.3.3
max-PRACH-FD-OccasionsPerSlot	Per configuration	3GPP TS 38.211, sec 6.3.3.2
maxNum-PRACH-Configurations	Max number of PRACH configurations	3GPP TS 38.331, 6.3.2
maxNum-TD-FD-PRACH-OccasionsPerSlot	Max number of Time and Frequency domain ROs per slot, across all configurations	Outside the scope of 3GPP
prach-RootProcessingRate	Number of roots that can be processed per unit of time. Can be per preamble format	Outside the scope of 3GPP
prach-OccasionsQueueSize	The maximum queue size for processing Rach Occasions	Outside the scope of 3GPP
maxNum-UCI-Maps	Limitations regarding to the storage of UCI Part1→Part2 maps: maximum number of maps	Outside the scope of 3GPP
maxNumCarriersBWLayersProduct-UL	Support for forming joint precoders across bandwidth, and across baseband antenna ports. Includes the option of “no support”.	Outside the scope of 3GPP
maxNum-MU-MIMO-LayersPer-UL-ChannelType	The maximum number of layers that can be multiplexed (can be channel specific)	Outside the scope of 3GPP
max-MIMO-LayersNon-CB-PUSCH max-MIMO-Layers-CB-PUSCH	The maximum number of layers that can be allocated to a UE, per MIMO scheduling type	3GPP TS 38.211, sec 6.3.1.3
...	...	...

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6.3.2b Summary of Configuration

A carrier to which the AAL_UPLINK_High-PHY channel models apply shall be configurable by the application.

Table 6.3.2b-1 lists a subset of the carrier configurations pertinent to AAL_UPLINK_High-PHY.

NOTE: Additional configuration parameters are possible for the AALI implementation.

Table 6.3.2b-1: Sample Carrier Configurations pertinent to AAL_UPLINK_High-PHY

Configuration	Short summary	3GPP Spec Reference
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ul-Bandwidth	Carrier bandwidth for the UL reception	3GPP TS 38.104, sec 5.2, 5.3 3GPP TS 38.211, sec 4.4.4.2
ul-Frequency	Carrier reference frequency (e.g. Point A, or RF reference frequency), for the UL reception	3GPP TS 38.104, sec 5.2, 5.3 3GPP TS 38.211, sec 4.4.4.2
ul-GridSize	numerology μ , for UL reception (except PRACH); can correspond to 15, 30, 60, 120 kHz.	3GPP TS 38.211, sec 4.2
ul-CarrierOffset	Offset to carrier or k_0^μ , per numerology μ	3GPP TS 38.331, sec 6.3.2 3GPP TS 38.211, sec 5.3.1
max-RX-Antennas	Number of baseband antenna ports for UL reception	Outside the scope of 3GPP
shift7dot5kHz	Indicates whether there is 7.5 kHz shift or not.	3GPP TS 38.331, sec 6.3.2
uci-Configuration	Semistatic maps for to reference in assisting L1 to computed the actual size of CSI part 2	3GPP TS 38.213, sec 9.3
rssi-MeasurementSupport rsrp-MeasurementSupport	Selection of power metrics (e.g. absolute, relative) and scale (dB, linear)	Partly outside the scope of 3GPP. Some of the units are defined in 3GPP TS 38.133, e.g. for RSSI and RSRP
beamformingTable	Semistatic configuration beamforming vectors for UL reception	Outside the scope of 3GPP
[Semistatic configuration of UL Channel Model parameters]	Alternative semistatic provisioning of some semistatic UL Channel Model parameters, e.g. see the respective PRACH parameters in the PRACH Chanel Model Section 6.3.6.2: PRACH: restrictedSetConfig, $N_{preamble}^{total}$, N_{cs} , M , k_1 , μ_{PUSCH} , μ_{PRACH} , $\bar{k}$, prachStartSymbol, prachFormat, $N_t^{RA,slot}$, preambleStartIndex	c.f Table 6.3.6.2-1
[U-plane configuration]	Semistatic configuration for correct routing of OFH packets towards the O-RU. The parameters include, but not limited to: - Processing Elements - Configured low-level-rx-end points - Receive carriers	Sec D.3.8 of ORAN-WG4-MP.0.v007.01

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2 NOTE 1: Additional configuration parameters may be added in future releases to support additional features beyond
3 basic 3GPP Rel-15.

NOTE 2: The summary description of configuration parameters outside the scope of 3GPP is not meant to restrict the mechanism or data type for such configuration.

6.3.3 PUSCH Channel Model

Per section 5.1.3.3.7 of the O-RAN AAL GAnP document [7] for the PUSCH High-PHY Profile, the PUSCH Channel model of the AAL_UPLINK_High-PHY Profile supports acceleration of PUSCH Data, DM-RS and PT-RS functionality.

The set of accelerated functions associated with the processing of PUSCH TB(s) is as follows:

- IQ decompression¹
- RE de-mapping
- Combining
- Channel estimation (see DM-RS and PT-RS)
- Channel equalization (see DM-RS and PT-RS)
- Transform precoding (optional- only required for DFT-s-OFDM waveform)
- Demodulation
- Descrambling
- Rate de-matching
- LDPC decoding
- CRC check
- UCI Decoding

The set of accelerated functions associated with the processing of PUSCH DM-RS is as follows:

- IQ decompression¹
- RE de-mapping
- Combining
- Demodulation
- Sequence detection

The set of accelerated functions associated with the processing of PUSCH PT-RS is as follows:

- IQ decompression¹
- RE de-mapping
- Combining
- Demodulation
- Layer demapping

- Sequence detection

6.3.3.1 PUSCH input and output for AAL_UPLINK_High-PHY Profile

The AAL_UPLINK_High-PHY profile shall signal PUSCH allocation(s) per slot.

From the AAL Application, the PUSCH interface receives the associated parameters for the PUSCH allocation.

The output data consists of:

- the CRC status for the PUSCH codeword, as well as optionally for individual CBs.
- in cases of successful CRC verification: the TB(s) output of the decoder.
- in case of UCI inclusion: the included UCI payload(s)
- in case of UL HARQ buffer release: asynchronous error indication with identifiers related to the corresponding released UL HARQ buffers in HWA

6.3.3.2 PUSCH Parameters

The following parameters are required to be supported by the AALI implementation when offloading operations. The AAL Application shall supply all relevant parameters; for ease of reading, the parameters are organized per signal type: Data, DM-RS, PT-RS.

PUSCH Data Parameters

Table 6.3.3.2-1: PUSCH Data Parameters

AAL-LPU parameters	Short summary	3GPP Spec Reference
MAC TB		
tb-Size	Transport block size. Computed at L2.	3GPP TS 38.214, sec 6.1.4.2
← mac-PDU (if TB CRC passes)	This is an output: the TB corresponding to the PUSCH codeword	
UCI detection (if relevant)		
csi-Part1BitLength	Size of UCI part 1 (or single UCI)	3GPP TS 38.212, sec 6.3.2.4
alphaScaling	Needed to compute number of coded symbols per layer	3GPP TS 38.212, sec 6.3.2.4
harq-ACK-BitLength	Number of HARQ-ACK bits	3GPP TS 38.212, sec 6.3.2.4
betaOffset-HARQ-ACK or codedModulationSymbPerLayer-ACK	betaOffsetHarq: Beta Offset for HARQ-ACK bits. codedModulationSymbPerLayer: $Q'_{\{\text{ACK}\}}$, the number of coded modulation symbols per layer for HARQ-ACK transmission	3GPP TS 38.212, sec 6.3.2.4
betaOffset-CSI-Part1 or codedModulationSymbPerLayer-CSI-Part1	betaOffsetCsiPart1: Beta Offset for CSI part 1 bits. codedModulationSymbPerLayer-CSI-Part1: $Q'_{\{\text{CSI-1}\}}$, number of	3GPP TS 38.212, sec 6.3.2.4

	coded modulation symbols per layer for CSI part 1 transmission	
betaOffset-CSI-Part2	Beta Offset for CSI part 2 bits.	3GPP TS 38.212, sec 6.3.2.4
uci-Part1ToPart2Correspondence	Mappings from Csi Part1 to length and priorities of CSI part 2 reports. Needed to compute in L1 the actual size of CSI part 2	3GPP TS 38.213, sec 9.3
← uci-Payload	This is an output: Uninterpreted UCI payload; part 1, if UCI is composed of two parts	
← uci-Part2Payload	This is an output: Uninterpreted UCI part 2, if UCI is composed of two parts	
CRC Check		
← crc-Indication	This is an output: CRC Status for CW + other metrics (e.g. SINR)	38.212, section 6.2.1
CB CRC and CB Desegmentation		
num-CB	number of expected code blocks	3GPP TS 38.214, sec 5.2.2
cb-PresentAndPosition[*]	Presence indicator (e.g. bitmap), for each CB	3GPP TS 38.214, sec 6.2.3
← cb-CRC-Status	This is an output: CRC Status per CB + other metrics (e.g. SINR)	3GPP TS 38.214, sec 6.2.3
LDPC Decoding		
rv-Index	redundancy version, per CW. Determines starting position k_0 in the circular buffer. In DCI, or from aggregation.	3GPP TS 38.214, sec 6.1.4, sec 6.1.2.1
targetCodeRate or (mcs-Table and mcs-Index)	targetCodeRate: R, target code rate Explicit, for initial transmission. Implicit from TBSize and allocation, for retransmissions. MCS Table and MCS index: target code rate can be extracted from the index (first transmission), or as above for retransmissions.	3GPP TS 38.214, sec 6.1.4.1
newData	Signals whether there is a new transmission, or a retransmission (per CW)	3GPP TS 38.212, sec 7.3.1.1 3GPP TS 38.214, sec 5.1.7.2
harq-Process-ID	harq process number for the UL buffer	3GPP TS 38.212, sec 7.3.1.1
ldpc-BaseGraph	LDPC base graph to use (can be explicit, or derived from TB size and initial target rate) (per CW)	3GPP TS 38.212, sec 7.2.2

numIterations	Number of iterations for LDPC decoding. Refer to NOTE 1 at bottom of table. Note: This is an optional parameter	Not applicable.
earlyTermination-OnOff	To enable/disable early termination of LDPC decoding. Refer to NOTE 2 at bottom of table. Note: This is an optional parameter	Not applicable.
Rate Dematching		
tb-Size-LBRM or n-CB	TBS_{LBRM}: Reference TBS for allocations subject to rateMatching = limitedBufferRM; impacts circular buffer length N_{CB} = circular buffer length, after account for any FBRM/LBRM considerations.	3GPP TS 38.212, sec 5.4.2.1, 6.2.5
num-CB	number of expected code blocks. Also uses the rate match size E_r, which is a computed field	3GPP TS 38.212, sec 6.2.6
cb-Size[*]	number of bits in each CB	3GPP TS 38.212, sec 6.2.6
CB Deconcatentation		
num-CB	number of expected code blocks. Also uses the rate match size E_r, which is a computed field	3GPP TS 38.212, sec 6.2.6
cb-Size[*]	number of bits in each CB	3GPP TS 38.212, sec 6.2.6
Descrambling		
n-ID-PUSCH or c-Init	n_{ID}: data scrambling identity (PCI by default) c_{init}: scrambling initialization	3GPP TS 38.211, sec 6.3.1.1
rnti	RNTI associated with the PDSCH transmission	3GPP TS 38.211, sec 6.3.1.1
Demodulation		
qam-ModOrder or (mcs-Table and mcs-Index)	Q_m: Signaled via MCS in DCI MCS Table and MCS index: modulation can be extracted from the MCS table and index.	3GPP TS 38.214, sec 6.1.4.1
IDFT for DFT-s-OFDM		
transformPrecoding	Signaled in DCI	3GPP TS 38.214, sec 6.3.1.4
Channel Estimation & Equalization		
<see PUSCH DMRS>		

<see PUSCH PTRS>			
Combining			
See section 6.2.7 and section 6.3.7		Combiner matrix is per PUSCH allocation	
RE Demapping			
Frequency Domain	bwp-Start	Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUSCH allocation	3GPP TS 38.211, sec 6.3.1.7
	bwp-Size	Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUSCH allocation	3GPP TS 38.211, sec 6.3.1.7
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	num-DMRS-CDM-GrpsNoData	No PUSCH mapping on DMRS CDM groups marked as having no data.	3GPP TS 38.212, sec 7.3.1.1
	resourceAlloc	Bitmap-based (type 0) or offset & length (type 1)	3GPP TS 38.214, sec 6.1.2.2
	rb-Bitmap	<u>type 0</u> : allocation is based on this bitmap. In 3GPP, it is signaled at RB group resolution	3GPP TS 38.214, sec 6.1.2.2.1 3GPP TS 38.212, sec 7.3.1.1.2
	rb-Start	<u>type 1</u> : start of allocation derived from DCI RIV	3GPP TS 38.214, sec 6.1.2.2.2 3GPP TS 38.212, sec 7.3.1.1.2
	rb-Size	<u>type 1</u> : number of continuously allocated VRBs derived from DCI RIV	3GPP TS 38.214, sec 5.1.2.2.2 3GPP TS 38.212, sec 7.3.1.1.2
	intraSlotFrequencyHopping	indicates whether PUSCH allocation is based on intra-slot frequency hopping	3GPP TS 38.212, sec 7.3.1.1 3GPP TS 38.214, sec 6.3
	pusch-SecondHop-PRB	location of the second frequency hop	3GPP TS 38.214, sec 6.3
	tx-DirectCurrentLocation	indicates the subcarrier index within the carrier corresponding to the numerology of the corresponding uplink BWP	3GPP TS 38.331, sec 6.3.2
	shift7dot5kHz	Indicates whether there is 7.5 kHz shift or not.	3GPP TS 38.331, sec 6.3.2
Time Domain	startSymbolIndex	Start symbol index, for the allocation	3GPP TS 38.214, Table 6.1.2.1-1
	nrOfSymbols	Number of symbols, for the allocation	3GPP TS 38.214, Table 6.1.2.1-1
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1

NOTE 1: As for numIterations parameter in LDPC Decoding, there can be several use cases to apply the optimal number of LDPC decoding iterations, such as- 1) in a case when more number of LDPC iterations (as part of the entire allowed processing resources) should be allocated to URLLC traffic than eMBB traffic if they coexist, 2) in a case when cell coverage would improve with increased number of LDPC iterations for cell-edge users (e.g., with minimum MCS and small PDU length), and 3) in a case when VoIP such as small data size would be benefited from increased number of LDPC iterations. The numIterations parameter in LDPC Decoding is assumed to be always set by AAL application as same as other parameters in LDPC Decoding, and the support of the numIterations parameter in AALI implementation at HWA is optional (e.g., the default value is “0”, which is set by AAL application when AALI implementation does not support this parameter. In that case, AALI implementation may simply choose to ignore this parameter (or the value)). Note that the aforementioned examples are not meant to be an exhaustive list, and AAL applications may choose to set this parameter for other use cases not mentioned here. The actual range of numIterations parameter value would be as per the AALI implementation.

NOTE 2: As for earlyTermination-OnOff parameter in LDPC Decoding, there would be several use cases to optimize the early termination of LDPC decoding iterations, such as – 1) in a case where it would be preferable to prioritize energy efficiency rather than minimizing the BLER (which leads to earlyTermination-OnOff to be on), and 2) in a case where early termination would be preferable for configured grant. The earlyTermination-OnOff parameter in LDPC Decoding is assumed to be always set by AAL application as same as other parameters in LDPC Decoding, and the support of the earlyTermination-OnOff parameter in AALI implementation at HWA is optional (e.g., the default value is “0”, which means early termination is off. AAL application will ONLY set this value to 1 (i.e. ON) when AALI implementation supports this feature.). Note that the aforementioned examples are not meant to be an exhaustive list, and AAL applications may choose to set this parameter for other use cases not mentioned here.

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2 PUSCH DM-RS Parameters

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Table 6.3.3.2-2: PUSCH DM-RS Parameters

AAL-LPU parameters	Short summary	3GPP Spec Reference
DMRS Sequence		
pusch-DMRS-ScramblingId or c-InitFirstSymbol	pusch-DMRS-ScramblingId: $N_{\{ID\}}$, scrambling identifier for $n_{\{SCID\}}$ c-InitFirstSymbol: $c_{\{init\}}[l]$, scrambling initialization for DMRS symbol 1 (CP-OFDM) or for entire DMRS (Rel- 15 DFT-S-OFDM)	3GPP TS 38.211, sec 6.4.1.1.1/2
n-SCID	sequence index (from DCI; defaults to 0)	3GPP TS 38.211, sec 6.4.1.1.1/2
pusch-DMRS-Identity	$N_{\{ID\}}^{\{RS\}}$, same as N_{ID}^{PUSCH} , or PCI (DFT-S-OFDM)	3GPP TS 38.211, sec 6.4.1.1.2
groupOrSequenceHopping	indicates whether sequence or group hopping is enabled	3GPP TS 38.211, sec 6.4.1.1.2
low-PAPR-GroupNumber	U: low PAPR group number [DFT-s- OFDM]	3GPP TS 38.211, sec 5.2.2
low-PAPR-SequenceNumber	V: low PAPR sequence number [DFT-s- OFDM]	3GPP TS 38.211, sec 5.2.2
Demodulation		

transformPrecoding		CP-OFDM: PUSCH DM-RS uses QPSK modulation DFT-s-OFDM: PUSCH DMRS uses low-PAPR sequences	3GPP TS 38.211, sec 6.4.1.1.1/2
Combining			
[see PUSCH data combining]		DMRS follows PUSCH-data combining	
ul-TPMI-Index		[optional] PMI of the UE, can help with combiner formation	3GPP TS 38.211, sec 6.3.1.5
RE Demapping			
Frequency Domain	bwp-Start	see PUSCH data table	3GPP TS 38.211, sec 6.3.1.7
	bwp-Size	see PUSCH data table	3GPP TS 38.211, sec 6.3.1.7
	subcarrierSpacing	see PUSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1
	dmrs-Ports	DMRS ports per layer; OC weights are applied to DM-RS REs, based on DM-RS CDM group and identifier for each DM-RS port.	3GPP TS 38.211, sec 6.4.1.1.3
Time Domain	ul-DMRS-SymbPos	location of DMRS in a slot sufficiently captures the impact of upper layer parameters, e.g. mapping type, additional pos	3GPP TS 38.211, sec 6.4.1.1.3
	dmrs-ConfigType	DMRS configuration type controls port to CDM group correspondence	3GPP TS 38.211, sec 6.4.1.1.3
	dmrs-Ports	DMRS ports per layer; OC weights are applied to DM-RS REs, based on DM-RS CDM group and identifier for each DM-RS port.	3GPP TS 38.211, sec 6.4.1.1.3
	cyclicPrefix	see PUSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
PUSCH extension information			
weightForAverageFO		To apply the weighted average frequency offset for channel estimation. Refer to NOTE 1 at bottom of the table. Note: This is an optional parameter.	Not applicable.

NOTE 1: As for weightForAverageFO parameter in PUSCH/PUCCH extension information, it is assumed that the weighted average value for the two FOs (Frequency Offset estimated by using DMRS at gNB) between the instantaneous FO and the current average FO would be applied individually for each slot in PUSCH/PUCCH, and AAL application would inform the HWA of the weight value (weightForAverageFO parameter) if necessary depending on events so that the HWA would decide how much the current average FO should be applied for better radio demodulation performance. The typical formula for the weighted average of frequency offset would be:

$$FO_ave_new = \mu * FO_ave_old + (1-\mu) * FO_instant$$

where

μ equals to the weightForAverageFO parameter,

FO_ave_new is the updated frequency offset value averaged for the slot,

FO_ave_old is the previous frequency offset value before update with the weightForAverageFO parameter,

$FO_instant$ is the instantaneous FO.

Examples of updating FO with the weightForAverageFO parameter would be:

- in a case that the additional DMRS is not configured, the AAL application guides HWA to use the current FO averaged for the slot ($\mu=1$),
- in a case that small number of RBs are allocated to a fast moving UE, the AAL application guides HWA how much the current average FO for the slot would be applied ($0 \leq \mu \leq 1$),
- in a case that the first UL data is scheduled after handover, the AAL application guides HWA to use the instantaneous FO for the slot ($\mu=0$; initializing the average FO).

Note that the aforementioned examples are not meant to be an exhaustive list, and AAL applications may choose to set this parameter for other use cases not mentioned here. The weightForAverageFO parameter in PUSCH/PUCCH extension information is assumed to be always set by AAL application as same as other parameters, and the support of the weightForAverageFO parameter in AALI implementation at HWA is optional (e.g., the default value is “0”, which is set by AAL application when AALI implementation does not support this parameter. In that case, AALI implementation may simply choose to ignore this parameter (or the value)).

PUSCH PT-RS Parameters

Table 6.3.3.2-3: PUSCH PT-RS Parameters

AAL-LPU parameters	Short summary	3GPP Spec Reference
Sequence Detection		
num-PTRS-Ports	number of PT-RS ports (Rel-15: only one)	3GPP TS 38.212, sec 7.3.1.1.2
ptrs-DMRS-port[*]	DM-RS ports associated with each of the PT_RS ports	3GPP TS 38.214, sec 6.2.3.1 3GPP TS 38.212, sec 7.3.1.1.2
ul-PTRS-SampleDensity	$N_{\text{group}}^{\text{PT-RS}}$: Number of PT-RS groups [DFTS-OFDM case]	3GPP TS 38.211, sec 6.4.1.2.1.2 3GPP TS 38.214, sec 6.2.3.2
ul-PTRS-TimeDensityTransformPrecoding	$N_{\text{samp}}^{\text{group}}$: Number of samples per PT-RS group [DFTS-OFDM case]	3GPP TS 38.211, sec 6.4.1.2.1.2 3GPP TS 38.214, sec 6.2.3.2
n-ID-PUSCH	N_{ID} : Identity for associated PUSCH	3GPP TS 38.211, sec 6.4.1.2.1.2
Layer Demapping		
num-PTRS-Ports	number of PT-RS ports	3GPP TS 38.212, sec 7.3.1.1.2

ptrs-DMRS-Port[*]		DM-RS ports associated with each of the PT-RS ports	3GPP TS 38.214, sec 6.2.3.1 3GPP TS 38.212, sec 7.3.1.1.2
Combining			
[same as the associated DMRS Port(s)]			
RE Demapping			
Frequency Domain	bwp-Start	see PUSCH data table	3GPP TS 38.211, section 7.3.1.6
	bwp-Size	see PUSCH data table	3GPP TS 38.211, section 7.3.1.6
	subcarrierSpacing	see PUSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1
	ptrs-FreqDensity	$K_{\{PT-RS\}}$, frequency density (includes the case where there is no PT-RS)	3GPP TS 38.214, sec 6.2.3.1, table 6.2.3.1-2
	ptrs-RE-Offset	$k_{\{ref\}}^{\{RE\}}$, re offset (can depend on DMRS association and RRC signaling of resourceElementOffset)	3GPP TS 38.211, sec 6.4.1.2.2, table 6.4.1.2.2.1-1 3GPP TS 38.331, sec 6.3.1
	rnti	RNTI associated with the PUSCH transmission	3GPP TS 38.211, sec 7.4.1.2.2
Time Domain	ptrs-TimeDensity	$L_{\{PT-RS\}}$, time density (includes the case where there is no PT-RS) [CP-OFDM]	3GPP TS 38.214, sec 6.2.3.1, table 6.2.3.1-1
	ul-DMRS-SymbPos	see PUSCH DMRS table symbols used for DMRS (PT-RS skips over these)	3GPP TS 38.211, sec 7.4.1.1.2 and Tables 7.4.1.1.2-3 and 7.4.1.1.2-4
	cyclicPrefix	see PUSCH data table	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
Power Offset			
	ul-PTRS-Power	$\alpha_{\{PTRS\}}^{\{PUSCH\}}$: PUSCH to PT-RS power ratio per layer per RE	3GPP TS 38.214, sec 6.2.3.1, Table 6.2.3.1-3

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6.3.4 PUCCH Channel Model

Per section 5.1.3.3.7 of the O-RAN AAL GAnP document [7] for the PUCCH High-PHY Profile, the PUCCH Channel model of the AAL_UPLINK_High-PHY Profile supports acceleration of PUCCH Format 0, Format 1 UCI and DM-RS and Formats 2,3,4 UCI and DM-RS.

The set of accelerated functions associated with the processing of PUCCH Format 0 is as follows:

- IQ decompression¹
- RE de-mapping
- Sequence detection

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The set of accelerated functions associated with the processing of PUCCH Format 1 UCI is as follows:

- IQ decompression¹

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- RE de-mapping
- Channel estimation
- Channel equalization
- Demodulation

The set of accelerated functions associated with the processing of PUCCH Format 1 DM-RS is as follows:

- IQ decompression¹
- RE de-mapping
- Combining
- Demodulation
- Sequence detection

The set of accelerated functions associated with the processing of PUCCH Formats 2, 3, 4 is as follows:

- IQ decompression¹
- RE de-mapping
- Channel estimation
- Channel equalization
- Transform precoding (optional- only required for DFT-s-OFDM waveform)
- Demodulation
- Descrambling
- Rate de-matching
- Polar/Block decoding
- CRC check

The set of accelerated functions associated with the processing of PUCCH Formats 2, 3, 4 DM-RS is as follows:

- IQ decompression¹
- RE de-mapping
- Combining
- Demodulation
- Sequence detection

6.3.4.1 PUCCH input and output for AAL_UPLINK_High-PHY Profile

The AAL_UPLINK_High-PHY profile shall signal PUCCH resource(s) per slot.

From the AAL Application, the PUCCH interface receives the associated parameters for the configuration of the PUCCH resource.

The output data consists of:

- the CRC or detection status of the PUCCH UCI payload (HARQ, SR or CSI), as well as related metrics (e.g. SINR)
- The UCI Payload: HARQ, SR (Formats 0/1) or transparent UCI payload(s) (Formats 2/3/4).

6.3.4.2 PUCCH Parameters

The following parameters are required to be supported by the AALI implementation when offloading operations. The AAL Application shall supply all relevant parameters; for ease of reading, the parameters are grouped as follows, in alignment with the O-RAN AAL GAnP document [7]: PUCCH Format 0, PUCCH Format 1 (UCI and DMRS), PUCCH Formats 2/3/4 (UCI and DMRS). Only one of these sets of parameters is applicable to any one PUCCH resource:

PUCCH Format 0 Parameters

Table 6.3.4.2-1: PUCCH Format 0 Parameters

AAL-LPU parameters		Short summary	3GPP Spec Reference
UCI			
sr-BitLen		indicates whether SR can be indicated (lengths 0 or 1)	3GPP TS 38.213, sec 9.2
bitLen-HARQ		indicates number of HARQ bits	3GPP TS 38.213, sec 9.2
← sr-Indication ← harq-Value[*]		This is an output: SR and HARQ observations, or outcome of detection; Also: metrics like SINR, RSRP, etc.	3GPP TS 38.213, sec 9.2
Sequence Detection			
pucch-GroupHopping		indicates whether group, sequence or no hopping is applied	3GPP TS 38.211, sec 6.3.2.2.1
n-ID-PUCCH-Hopping		hopping identifier	3GPP TS 38.211, sec 6.3.2.2.1
initialCyclicShift		initial cyclic shift	3GPP TS 38.211, sec 6.3.2.2.2
RE Demapping			
Frequency Domain	bwp-Start	Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7
	bwp-Size	Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	prb-Start	RB_{BWP}^{offset}, PRB offset, prior to any hopping	3GPP TS 38.213, sec 9.2.1

	intraSlotFrequencyHopping	indicates whether the allocation hops in frequency or not	3GPP TS 38.211, sec 6.3.2.2.1
	secondHop-PRB	RB offset of the second hop, in case of intra-slot frequency hopping	3GPP TS 38.213, sec 9.2.1
Time Domain	startSymbolIndex	first symbol for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
	nrOfSymbols	number of symbols for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1

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2 PUCCH Format 1 UCI Parameters

3 **Table 6.3.4.2-2: PUCCH Format 1 UCI Parameters**

AAL-LPU parameters		Short summary	3GPP Spec Reference
UCI			
sr-BitLen		indicates whether SR can be indicated (lengths 0 or 1)	3GPP TS 38.213, sec 9.2.4
bitLen-HARQ		indicates number of HARQ bits	3GPP TS 38.213, sec 9.2.2
← sr-Indication ← harq-Value[*]		This is an output: SR and HARQ observations, if any (or outcome of detection + metrics like SINR, RSRP)	
Demodulation			
pucch-GroupHopping		indicates whether group, sequence or no hopping is applied	3GPP TS 38.211, sec 6.3.2.2.1
n-ID-PUCCH-Hopping		hopping identifier	3GPP TS 38.211, sec 6.3.2.2.1
initialCyclicShift		initial cyclic shift	3GPP TS 38.211, sec 6.3.2.2.2
timeDomain-OCC-Idx		index of orthogonal sequence w	3GPP TS 38.211, sec 6.3.2.4.1 3GPP TS 38.213, section 9.2.1
sr-BitLen and bitLen-HARQ		Number of bits determines modulation: BPSK or QPSK	3GPP TS 38.211, sec 6.3.2.4.1
Channel Estimation and Equalization			
[see PUCCH Format 1 DM-RS]			
RE Demapping			
Frequency	bwp-Start	Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7

	bwp-Size	Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	prb-Start	$RB_{\{BWP\}^{\{offset\}}}$: PRB offset, prior to any hopping	3GPP TS 38.213, sec 9.2.1
	intraSlotFrequencyHopping	indicates whether the allocation hops in frequency or not	3GPP TS 38.211, sec 6.3.2.2.1
	secondHop-PRB	RB offset of the second hop, in case of intra-slot frequency hopping	3GPP TS 38.213, sec 9.2.1
Time Domain	startSymbolIndex	first symbol for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
	nrOfSymbols	number of symbols for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1

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2 PUCCH Format 1 DM-RS Parameters

3 **Table 6.3.4.2-3: PUCCH Format 1 DM-RS Parameters²**

AAL-LPU parameters		Short summary	3GPP Spec Reference
Descrambling			
pucch-GroupHopping		indicates whether group, sequence or no hopping is applied	3GPP TS 38.211, sec 6.3.2.2.1, 6.4.1.3.1.1
n-ID-PUCCH-Hopping		hopping identifier	3GPP TS 38.211, sec 6.3.2.2.1, 6.4.1.3.1.1
initialCyclicShift		initial cyclic shift	3GPP TS 38.211, sec 6.3.2.2.2, 6.4.1.3.1.1
timeDomain-OCC-Idx		index of orthogonal sequence w	3GPP TS 38.211, sec 6.3.2.4.1, 6.4.1.3.1.1 3GPP TS 38.213, sec 9.2.1
RE Demapping			
Frequency Domain	bwp-Start	Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7, 6.4.1.3.1.2

² All the parameters in this PUCCH Format 1 DM-RS table are also present in the PUCCH Format 1 UCI table

	bwp-Size	Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7, 6.4.1.3.1.2
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1, 6.4.1.3.1.2
	prb-Start	PRB offset, prior to any hopping	3GPP TS 38.213, sec 9.2.1 3GPP TS 38.211, sec 6.4.1.3.1.2
	intraSlotFrequencyHopping	indicates whether the allocation hops in frequency or not	3GPP TS 38.211, sec 6.3.2.2.1, 6.4.1.3.1.2
	secondHop-PRB	RB offset of the second hop, in case of intra-slot frequency hopping	3GPP TS 38.213, sec 9.2.1 3GPP TS 38.211, sec 6.4.1.3.1.2
Time Domain	startSymbolIndex	first symbol for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2 3GPP TS 38.211, sec 6.4.1.3.1.2
	nrOfSymbols	number of symbols for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2 3GPP TS 38.211, sec 6.4.1.3.1.2
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1, 6.4.1.3.1.2 3GPP TS 38.214, Table 5.1.2.1-1
PUCCH extension information			
weightForAverageFO		See PUSCH extension information in Table 6.3.3.2-2. Note: This is an optional parameter.	Not applicable.

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PUCCH Format 2/3/4 UCI Parameters

Table 6.3.4.2-4: PUCCH Formats 2/3/4 Parameters

AAL-LPU parameters	Short summary	3GPP Spec Reference
UCI		
sr-BitLen	$O^{\{SR\}}$, indicates the number of SR bits	3GPP TS 38.213, sec 9.2 3GPP TS 38.212, Table 6.3.1.4-1
bitLen-HARQ	$O^{\{ACK\}}$, indicates number of HARQ bits	3GPP TS 38.213, sec 9.2 3GPP TS 38.212, Table 6.3.1.4-1
csi-Part1BitLength	$O^{\{CSI-part1\}}$, indicates the number of CSI part 1 bits	3GPP TS 38.213, sec 9.2 3GPP TS 38.212, Table 6.3.1.4-1

uci-Part1ToPart2Correspondence	Mappings from CSI Part1 to length and priorities of CSI part 2 reports. Needed to compute in L1 the actual size of CSI part 1 (Formats 3,4)	3GPP TS 38.213, sec 9.2.5
← uci-Payload	This is an output: Uninterpreted UCI payload (or outcome of decoding + metrics like SINR, RSRP, etc.)	3GPP TS 38.213, sec 9.2
← uci-Part2Payload	This is an output: Uninterpreted UCI part 2, if UCI is composed of two parts	
formatType	2, 3 or 4	3GPP TS 38.213, sec 9.2.2
CRC check		
<acceleration based on hardcoded parameters>	CRC size and generator depend on the payload size A, and fixed parameters	3GPP TS 38.212, sec 6.3.1.2
Polar/Block decoding		
<acceleration based on hardcoded parameters>	Encoding depends on the payload size A, and corresponding number of code blocks, and fixed parameters	3GPP TS 38.212, sec 6.3.1.3
Rate de-matching		
nrOfSymbols	$N_{\text{symb,UCI}}^{\text{PUCCH,*}}$: number of symbols carrying UCI for PUCCH formats 2/3/4 respectively	3GPP TS 38.212, Table 6.3.1.4-1
pre-DFT-OCC-Idx	$N_{\text{SF}}^{\text{PUCCH,*}}$: spreading factor for PUCCH formats 4	3GPP TS 38.212, Table 6.3.1.4
prb-Size	$N_{\text{PRBI}}^{\text{PUCCH,*}}$: actual number of RBs used by the UE for this allocation, for each format, respectively	3GPP TS 38.213, sec 9.2.1
pi2-BPSK	$\pi/2$ -BPSK: indicates that Formats 3-4 use $\pi/2$ -BPSK, rather than QPSK modulation NOTE: QPSK for Format 2	3GPP TS 38.211, sec 6.3.2.6.2, sec 6.3.2.5.2
maxCodeRate	$R_{\text{UCI}}^{\text{max}}$: Max coding rate to determine how to feedback UCI	3GPP TS 38.212, Table 6.3.1.4.1-1
Descrambling		
rnti	RNTI associated with the PUCCH transmission	3GPP TS 38.211, sec 6.3.2.5.1 and 6.3.2.6.1
nId-PUCCH-Scrambling or c-Init	n_{ID} : scrambling id (PCI by default) c_{init} : scrambling initialization	3GPP TS 38.211, sec 6.3.2.5.1 and 6.3.2.6.1
Demodulation		

<acceleration based on already-defined parameters>		See $\pi/2$ -BPSK parameter	
Block-wise Despreading			
pre-DFT-OCC-Idx		index of orthogonal sequence for Format 4.	3GPP TS 38.213, section 9.2.1
pre-DFT-OCC-Len		length of orthogonal sequence for Format 4.	3GPP TS 38.213, section 9.2.1
IDFT for DFT-s-OFDM			
prb-Size		Actual number of RBs used by the UE for this allocation (Format 3)	3GPP TS 38.213, sec 9.2.1
number of symbols		number of symbols for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
Channel Estimation and Equalization			
[see PUCCH Format 2/3/4 DM-RS]			
RE Demapping			
Frequency Domain	bwp-Start	Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7
	bwp-Size	Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	prb-Start	PRB offset, prior to any hopping	3GPP TS 38.213, sec 9.2.1
	prb-Size	$N_{\text{PRB}}^{\text{PUCCH,*}}$: actual number of RBs used by the UE for this allocation, for each format, respectively	3GPP TS 38.213, sec 9.2.1
	intraSlotFrequencyHopping	indicates whether the allocation hops in frequency or not	3GPP TS 38.211, sec 6.3.2.2.1 3GPP TS 38.212, sec 6.3.1.4
	secondHop-PRB	RB offset of the second hop, in case of intra-slot frequency hopping	3GPP TS 38.213, sec 9.2.1
Time Domain	startSymbolIndex	first symbol for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
	nrOfSymbols	$N_{\text{ymb,UCI}}^{\text{PUCCH,*}}$: number of symbols for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1

1 PUCCH Format 2/3/4 DM-RS Parameters

2 **Table 6.3.4.2-5: PUCCH Formats 2/3/4 DM-RS Parameters**

AAL-LPU parameters		Short summary	3GPP Spec Reference
Descrambling			
nId0-PUCCH-DMRS-Scrambling or c-InitFirstSymbol		$N_{\{ID\}}^0$: scrambling ID 0 for Format 2	3GPP TS 38.211, sec 6.4.1.3.2.1
		$c_{\{init\}}[l]$: scrambling initialization, per DMRS symb l.	
pucch-GroupHopping		indicates whether group, sequence or no hopping is applied (Formats 3/4)	3GPP TS 38.211, sec 6.3.2.2.1
n-ID-PUCCH-Hopping		hopping identifier (Formats 3/4)	3GPP TS 38.211, sec 6.3.2.2.1
initialCyclicShift		initial cyclic shift (Formats 3/4)	3GPP TS 38.211, sec 6.3.2.2.2
pre-DFT-OCC-Idx		index of orthogonal sequence for Format 4.	3GPP TS 38.213, section 9.2.1
pre-DFT-OCC-Len		length of orthogonal sequence for Format 4.	3GPP TS 38.213, section 9.2.1
RE Demapping			
Frequency Domain	bwp-Start	Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7
	bwp-Size	Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the PUCCH allocation	3GPP TS 38.211, sec 6.3.1.7
	subcarrierSpacing	subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	prb-Start	PRB offset, prior to any hopping	3GPP TS 38.213, sec 9.2.1
	prb-Size	$N_{\{PRB\}}^{\{PUCCH,*\}}$: Actual number of RBs used by the UE for this allocation, for each format, respectively	3GPP TS 38.213, sec 9.2.1
	intraSlotFrequencyHopping	indicates whether the allocation hops in frequency or not	3GPP TS 38.211, sec 6.3.2.2.1 3GPP TS 38.212, sec 6.3.1.4
	secondHop-PRB	RB offset of the second hop, in case of intra-slot frequency hopping	3GPP TS 38.213, sec 9.2.1
Time Domain	startSymbolIndex	first symbol for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2

	nrOfSymbols	$N_{\text{symb,UCI}}^{\text{PUCCH,*}}$: number of symbols for the PUCCH allocation	3GPP TS 38.213, sec 9.2.2
	add-DMRS-Flag	Indicates 2 additional DMRS symbols per hop of a PUCCH Format 3 or 4 (depending on length)	3GPP TS 38.213, sec 9.2.2
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1
PUCCH extension information			
	weightForAverageFO	See PUSCH extension information in Table 6.3.3.2-2. Note: This is an optional parameter.	Not applicable.

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6.3.5 SRS Channel Model

Per section 5.1.3.3.7 of the O-RAN AAL GAnP document [7] for the SRS High-PHY Profile, the SRS Channel model of the AAL_UPLINK_High-PHY Profile supports acceleration of SRS.

The set of accelerated functions associated with the processing of SRS is as follows:

- IQ decompression¹
- RE de-mapping
- Channel estimation

6.3.5.1 SRS input and output for AAL_UPLINK_High-PHY Profile

The AAL_UPLINK_High-PHY profile shall signal SRS Resource(s) per slot.

From the AAL Application, the SRS interface receives the associated parameters for the configuration of the SRS Resource.

The output data consists of the channel estimation metrics

6.3.5.2 SRS Parameters

The following parameters are required to be supported by the AALI implementation when offloading operations. The AAL Application shall supply all relevant parameters; for ease of reading, the parameters are grouped as follows, in alignment with the O-RAN AAL GAnP document [7]:

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19 SRS Parameters

20 **Table 6.3.5.2-1: SRS Parameters**

AAL-LPU parameters	Short summary	3GPP Spec Reference

Channel Estimation (report types are per implementation and may be optional. Combinations - of one or more, or even none - of report types linked to an SRS resource are also possible, and AAL-LPU shall indicate this in a capability)			
srs-Usage		usage of the SRS resource: beamManagement, codebook, nonCodebook, antennaSwitching	3GPP TS 38.331, sec 6.3.2
report type		Requested report type; may depend on usage	
SVD report type	iq-Representation		Size and formats e.g. BFP, etc used for encoding channel eigenvector matrices (format of U, V entries)
	singularValueRepresentation		Size and formats e.g. BFP, etc used for encoding channel singular values
	prg-Size		frequency resolution of SVD report, e.g. RB, PRG, etc
	← channel-SVD-Representation		at the required resolution
H (raw channel) report type	iq-Representation		Size and formats e.g. BFP, etc used for encoding channel matrices (format of H entries)
	prg-Size		frequency resolution of H report, e.g. RB, PRG, etc
	← channel-IQ-Matrix		at the required resolution
SINR report type	prg-Size		frequency resolution of SINR report, e.g. RB, PRG, etc
	← sinr-Report		at the required resolution (per symbol)
2D-DFT basis report type	prg-Size		Frequency resolution of 2D-DFT reports, e.g. RB, PRG, etc
	← channel-2D-DFT-Representation	beam-Indices	Strongest K_DFT_beams 2D-DFT beam indices per SRS basis vector
		beam-Values	Measured beam values for K_DFT_beams strongest 2D-DFT beams per SRS basis vector
		Rank	Channel rank

Additional report types can be added				
RE demapping				
numAntPorts		N_{ap}^{\{SRS\}}: number of SRS Ports		3GPP TS 38.211, sec 6.4.1.4.1
cyclicShift		n_{SRS}^{\{CS\}}: cyclic shift		3GPP TS 38.211, sec 6.4.1.4.2
sequence-ID		n_{ID}^{\{SRS\}}: SRS sequence identity		3GPP TS 38.211, sec 6.4.1.4.2
groupOrSequenceHopping		SRS Group and Sequence Hopping: Group, sequence or neither		3GPP TS 38.211, sec 6.4.1.4.2
Frequency domain	bwp-Start		Start, w.r.t. CRB, of bandwidth part: VRBs indexing is relative to the Bandwidth part for the allocation	3GPP TS 38.211, sec 6.3.1.7
	bwp-Size		Size of bandwidth part: VRBs indexing is relative to the Bandwidth part for the allocation	3GPP TS 38.211, sec 6.3.1.7
	subcarrierSpacing		subcarrier spacing impacts waveform generation, including centering	3GPP TS 38.211, sec 4.2, 5.3.1
	combSize		K_{TC}: transmission comb number	3GPP TS 38.211, sec 6.4.1.4.2
	combOffset		\bar{k}_{TC}: transmission comb offset	3GPP TS 38.211, sec 6.4.1.4.3
	frequencyShift		n_{shift}: frequency domain shift value	3GPP TS 38.211, sec 6.4.1.4.3
	frequencyPosition		n_{RRC}: frequency domain position	3GPP TS 38.211, sec 6.4.1.4.3
	Frequency Hopping Signaling Alt-1: RRC parameters	configIndex	C_{SRS}: SRS bandwidth config index	3GPP TS 38.211, sec 6.4.1.4.3
		bandwidthIndex	B_{SRS}: SRS bandwidth index	3GPP TS 38.211, sec 6.4.1.4.3
		frequencyHopping	b_{hop}: frequency hopping	3GPP TS 38.211, sec 6.4.1.4.3
	Frequency Hopping Signaling Alt-2: L2 computed fields	srs-BandwidthSize	m_{SRS,b}: SRS bandwidth size	3GPP TS 38.211, sec 6.4.1.4.3
		srs-BandwidthStart	PRB index (w.r.t. CRB0) for the start of SRS signal transmission, per symbol	3GPP TS 38.211, section 6.4.1.4.2

Time domain	numSymbols	$N_{\text{symb}}^{\text{SRS}}$: number of consecutive OFDM symbols for SRS	3GPP TS 38.211, sec 6.4.1.4.1
	timeStartPosition	l_0 : the starting position in the time domain given	3GPP TS 38.211, sec 6.4.1.4.1
	t-SRS	T_{SRS} : periodicity for SRS	3GPP TS 38.211, sec 6.4.1.4.3
	t-Offset	T_{offset} : slot offset	3GPP TS 38.211, sec 6.4.1.4.3
	numRepetitions	Corresponds to R: Repetition factor	3GPP TS 38.211, sec 6.4.1.4.3

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6.3.6 PRACH Channel Model

Per section 5.1.3.3.7 of the O-RAN AAL GAnP document [7] for the PRACH High-PHY Profile, the PRACH Channel model of the AAL_UPLINK_High-PHY Profile supports acceleration of PRACH.

The set of accelerated functions associated with the processing of PRACH is as follows:

- IQ decompression¹
- RE de-mapping
- Root sequence generation and correlation
- IFFT
- Noise estimation
- Peak search for power delay profile
- Preamble detection and delay/timing advance estimation

6.3.6.1 PRACH input and output for AAL_UPLINK_High-PHY Profile

The AAL_UPLINK_High-PHY profile shall signal PRACH Resource Occasion(s) per slot.

From the AAL Application, the PRACH interface receives the associated parameters for the configuration of the PRACH Occasion.

The output data consists of the PRACH detection outcome and metrics

6.3.6.2 PRACH Parameters

The following parameters are required to be supported by the AALI implementation when offloading operations. The AAL Application shall supply all relevant parameters; for ease of reading, the parameters are grouped as follows, in alignment with the O-RAN AAL GAnP document [7]:

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PRACH Parameters

Table 6.3.6.2-1: PRACH Parameters

AAL-LPU parameters	Short summary	3GPP Spec Reference
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Preamble Detection + Delay Estimation		
← preambleIndex[*]	list of detected preambles, per RO	
← timingAdvanceOffset[*]	per detected preamble & RO: timing advance	3GPP TS 38.213, sec 4.2
← detection metrics: - preamblePwr[*] - preamble-SNR[*]	per detected preamble & RO: SINR, Rx Power, etc.	
Peak Search		
<see Root Sequence Correlation>	Correlation is over the root sequences with valid preambles in each RACH occasion. The mechanism for detecting peaks is per implementation.	
Noise Estimation		
<per implementation>	One possibility is to perform noise estimation on an unused root sequence, in which case the identity of the unused root sequence can be signaled. NOTE: Noise estimation is implementation-specific, and this channel profile does not currently assume any one implementation.	
iFFT		
<based on RE mapping parameters and sequence length>	The formula for generating the PRACH waveform is provided in the reference to the right.	3GPP TS 38.211, sec 5.3.2
Root Sequence Correlation		
restrictedSetConfig	Configuration of an unrestricted set or one of two types of restricted sets, type A and type B	3GPP TS 38.211, sec 6.3.3.1
numPreambleIndices	$N_{\text{preamble}}^{\text{total}}$: number of preambles, per PRACH occasions	3GPP TS 38.213, section 8.1
startPreambleIndex	preamble index start for each PRACH occasion in this PRACH configuration	3GPP TS 38.213, section 8.1
prach-ZeroCorrConf	PRACH Zero CorrelationZone Config which is used N_{cs} : cyclic shift interval	3GPP TS 38.211, sec 6.3.3.1
RE Demapping		

Frequency Domain	num-FD-RA	From M: number of FDM occasions per PRACH slot	3GPP TS 38.211, sec 6.3.3.2
	prach-FrequencyOffset	$k1[*]$: offset in units of BWP REs (per RO)	3GPP TS 38.211, sec 5.3.2
	ul-BWP-PUSCH-SCS	μ_{PUSCH} : subcarrier spacing of UL BWP and of maximal SCS BWP	3GPP TS 38.211, sec 5.3.2
	prach-SubcarrierSpacing	μ_{PRACH} : subcarrier spacing of PRACH preamble	3GPP TS 38.211, sec 5.3.2
	prach-GuardbandOffset	$\bar{k}$: guard offset from $k1$ (can be derived from the SCS for PRACH and PUSCH and PRACH format)	3GPP TS 38.211, sec 6.3.3
Time Domain	prach-StartSymbol	starting symbol	3GPP TS 38.211, sec 6.3.3.2, Tables 6.3.3.2-2 to 6.3.3.2-4
	prach-Format	formats of the PRACH slot (0, 1, 2, 3, A1, ... A3/B3)	3GPP TS 38.213, sec 9.2.2
	numPRACH-Ocas	From $N_{t^{\text{RA,slot}}}$: number of Time domain PRACH occasions within a PRACH slot	3GPP TS 38.213, sec 9.2.2
	cyclicPrefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	3GPP TS 38.211, sec 4.2, 5.3.1 3GPP TS 38.214, Table 5.1.2.1-1

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6.3.7 Beamforming

In this release, the AAL_UPLINK_High-PHY profile supports the following beamforming methods available for OFH signaling:

- Predefined-Beam Beamforming, as defined in section 10.4.2.1 of [8];
- Weight-based Dynamic Beamforming, as defined in section 10.4.2.2 of [8];
- Attribute-Based Dynamic Beamforming, as defined in section 10.4.3 of [8];
- Channel-Information-Based Beamforming, as defined in section 10.4.4 of [8].

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6.3.7.1 Predefined-Beam Beamforming

The AALI shall support the AAL Application signaling as described in section 6.2.7.1

6.3.7.2 Weight-based Dynamic Beamforming

The AALI shall support the AAL Application signaling of the following parameters, as needed for AAL_UPLINK_High-PHY profile signaling of beamforming weight vectors of (b_{fw}I, b_{fw}Q) in C-Plane Extensions 1, 11, 19 [8]:

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Table 6.3.7.2-1: Weight-based Dynamic Beamforming Parameters for Uplink

AAL-LPU Parameters	Summary	AAL-LPU Role
AAL-LPU → AAL Application		
Channel Estimation Abstraction	A representation of SRS-based channel observations, as documented in the SRS channel model in section 6.3.5.2, e.g. for usage: codebook or usage: noncodebook	Signals an abstraction of the Channel Estimates to the AAL Application
AAL Application → AAL-LPU		
rnti [*]	Selected UEs for scheduling (for an illustration refer to the L users referenced in Appendix J.4 of [8])	Compute combining weights for the selected UEs and Precoders, e.g. based on the reported Channel Estimation and its Abstraction. Signal to O-RU beamforming vectors of weights (bfwI, bfwQ) over C-Plane for the AAL Application-selected UEs and Precoders
ul-TPMI	UE-based Precoder: selected precoding matrix for scheduling the UEs (see section 6.3.1.5 in of 3GPP TS 38.211)	

The High-PHY AAL API shall be extensible to allow, in future releases, the optional ability for the beamforming weights to be generated outside the AAL-LPU that consumes them for generating the appropriate C- and U-plane signaling by the accelerator, e.g. to generate the corresponding U-Plane eAxC I/Q sample streams and C-plane uelId field(s).

6.3.7.3 Attribute-Based Dynamic Beamforming

The AALI shall support the AAL Application signaling as described in section 6.2.7.3

6.3.7.4 Channel-Information-Based Beamforming

The AALI shall support the AAL Application signaling of the following parameters, as needed for AAL-LPU-signaling of channel estimate vectors of (ciIsample, ciQsample) in C-Plane Section 6 [8]:

Table 6.3.7.4-1: Channel-Information-Based Beamforming Parameters for Downlink

AAL-LPU Parameters	Summary	AAL-LPU Role
AAL-LPU → AAL Application		
[Channel Estimation Abstraction]	A representation of SRS-based channel observations, as documented in the SRS channel model in section 6.3.5.2, e.g. for usage: codebook or usage: noncodebook	Signals an abstraction of the Channel Estimation the AAL Application
AAL Application → AAL-LPU		
rnti	Selected UEs for scheduling (for an illustration refer to the L users referenced in Appendix J.4 of [8])	Signal to O-RU the Channel Estimates for the AAL Application-selected UEs and Precoders

chosenLayers[*]	Per UE: Selected precoding matrix for scheduling the UEs (see section 6.3.1.5 in of 3GPP TS 38.211)	
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Annex A [informative] Stage-3 Support for AAL_DOWNLINK_High-PHY and AAL_UPLINK_High-PHY using FAPI

This annex illustrates how SCF-222 [i.1] and SCF-223 [i.2] can be used to provide stage-3 support for the AAL_DOWNLINK_High-PHY and AAL_UPLINK_High-PHY profiles in this document.

In particular:

- Profile capabilities (sections 6.2.2 and 6.3.2) and Configurations (sections 6.2.2b and 6.3.2b) are supported via the SCF P5 interface
- Profile slot-level APIs (DL sections 6.2.3 – 6.2.7, UL sections 6.3.3 – 6.3.7) are supported via the SCF P7 interface
- Channel Models referenced in this document correspond to PDUs, which are transported over SCF P7 APIs in SCF-222

NOTE:

- this illustration does not preclude other stage-3 implementations of the the stage-2 APIs referenced this document, nor does it preclude the addition of other features or extensions in the future.
- alignment of this document with non-O-RAN Alliance specifications is not required.

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An illustration of this stage-3 support is further illustrated below for mapping of the CSI-RS Channel Model parameters from section 6.2.5 to CSI-RS PDU and related parameters in SCF-222 [i.1] and SCF-223 [i.2]:

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AAL-LPU Parameters		Summary	SCF-222 [i.1] / SCF-223 [i.2] Reference [i.1]
Sequence Generation			
n_{ID} or c_{init}[1]		n_{ID}: scrambling id c_{init}[1]: scrambling initialization for symb [1]	ScrambId signals n_{ID}
Modulation			
<acceleration based on hard-coded values>		CSI-RS uses QPSK modulation	<QPSK>, implicit
Precoding			
See section 6.2.7		Conceptually similar to FH signaling when precoding for Cat-B. 3GPP leaves DL precoding to implementation.	Precoding and Beamforming Attribute-based beamforming is supported in SCF-223
RE Mapping			
Frequency Domain	$\mu, \mu_{\{0\}}$	subcarrier spacing impacts waveform generation, including centering	SubcarrierSpacing signals μ

			The full set of numerologies, including $\mu_{\{0\}}$ is semistatically configured in P5 <i>dlFrequency</i> .
	startingRB	PRB where this CSI resource starts in relation to CRB#0	<i>StartRB</i> signals startingRB
	nrofRBs	Number of PRBs across which this CSI resource spans.	<i>NrofRBs</i> signals nrofRBs
	Frequency Domain Allocation	Bitmap defining the frequencyDomainAllocation, with interpretation subject to the Row selection for table 7.4.1.5.3-1	<i>FreqDomain</i> signals Frequency Domain Allocation
	CSI-RS locations Row	row indicating the CSI-RS location in table 7.4.1.5.3-1. Can be used to derive; - density ρ (see also Frequency Density) - cdmType - ports $\backslash\text{bar}\{k\}$ in the $(\backslash\text{bar}\{k\}, \backslash\text{bar}\{1\})$-tuple	<i>Row</i> signals CSI-RS locations Row
Time Domain	Density Dot5 Prb Location	Indicates whether even or odd PRBs are occupied by CSI-RS. Applicable when density = dot5 (0.5)	<i>FreqDensity</i> signals Density Dot5 Prb Location
	CSI-RS locations Row	row indicating the CSI-RS location in table 7.4.1.5.3-1. Can be used to derive; $\backslash\text{bar}\{1\}$ in the $(\backslash\text{bar}\{k\}, \backslash\text{bar}\{1\})$-tuple	<i>Row</i> signals CSI-RS locations Row
	l_0	value of l_0 for the interpretation of table 7.4.1.5.3-1. Signaled by RRC parameter firstOFDMSymbolInTimeDomain	<i>SymbL0</i> signals l_0
	l_1	value of l_1 for the interpretation of table 7.4.1.5.3-1. Signaled by RRC parameter firstOFDMSymbolInTimeDomain2	<i>SymbL1</i> signals l_1
	Cyclic Prefix	Cyclic prefix type. Impacts time-domain allocation, including waveform generation	<i>CyclicPrefix</i> signals Cyclic Prefix
Power Offset			
	[Power offset w.r.t. SSS]	power offset with respect to SSS	<i>powerControlOffsetProfileNR</i> and <i>powerControlOffsetSSProfileNR</i> signal power offset via 3GPP look-up tables.

		<i>csiRsPowerOffsetProfileSSS</i> signals actual “Ratio of CSI-RS EPRE to SSS EPRE” [i.1].
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Annex (informative): Change history/Change request (history)

Date	Revision	Description
2025.03.14	08.00	Placement of NOTE.1 and NOTE.2 with Table 6.3.3.2-1 is changed from bottom of Table 6.3.3.2-2 to bottom of Table 6.3.2.2-1. Addition of weightForAverageFO parameter for channel estimation in PUSCH/PUCCH (clauses 6.3.3.2, 6.3.4.2). Addition of fault notification in PDSCH (clause 6.2.3.1), PDCCH (clause 6.2.4.1) and PUSCH (clause 6.3.3.1).
2024.12.10	07.00	Addition of new parameters for LDPC decoding in PUSCH Data parameters (clause 6.3.3.2).
2024.03.25	06.00	Published from version 05.00.02
2023.03.17	05.00	Published from version 04.00.02
2022.11.21	04.00	Published from version 03.00.04
2022.07.29	03.00	Published from version 02.00.03
2022.03.29	02.00	Published from version 01.00.01
2021.11.15	01.00	Published from version 00.00.12